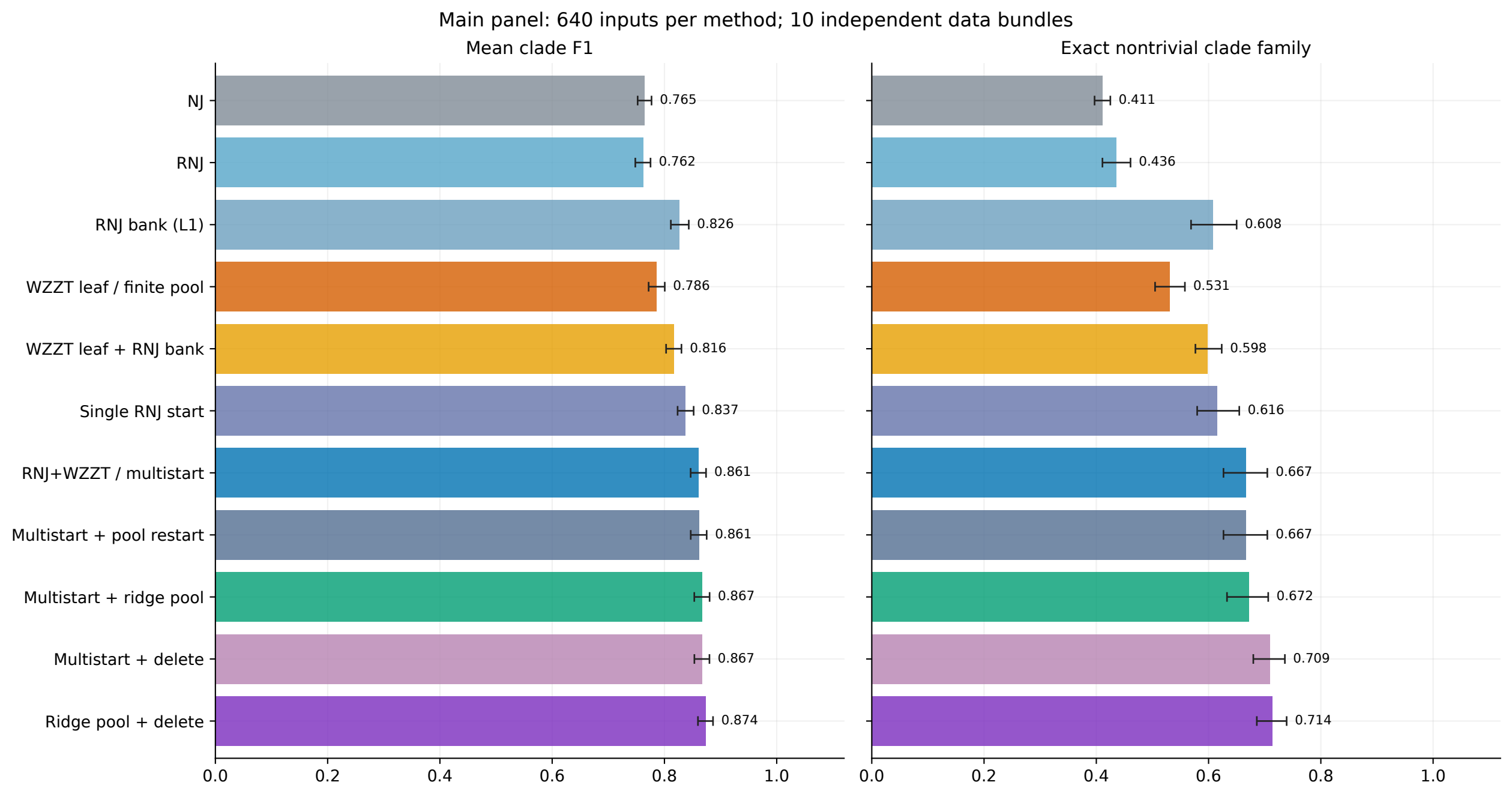
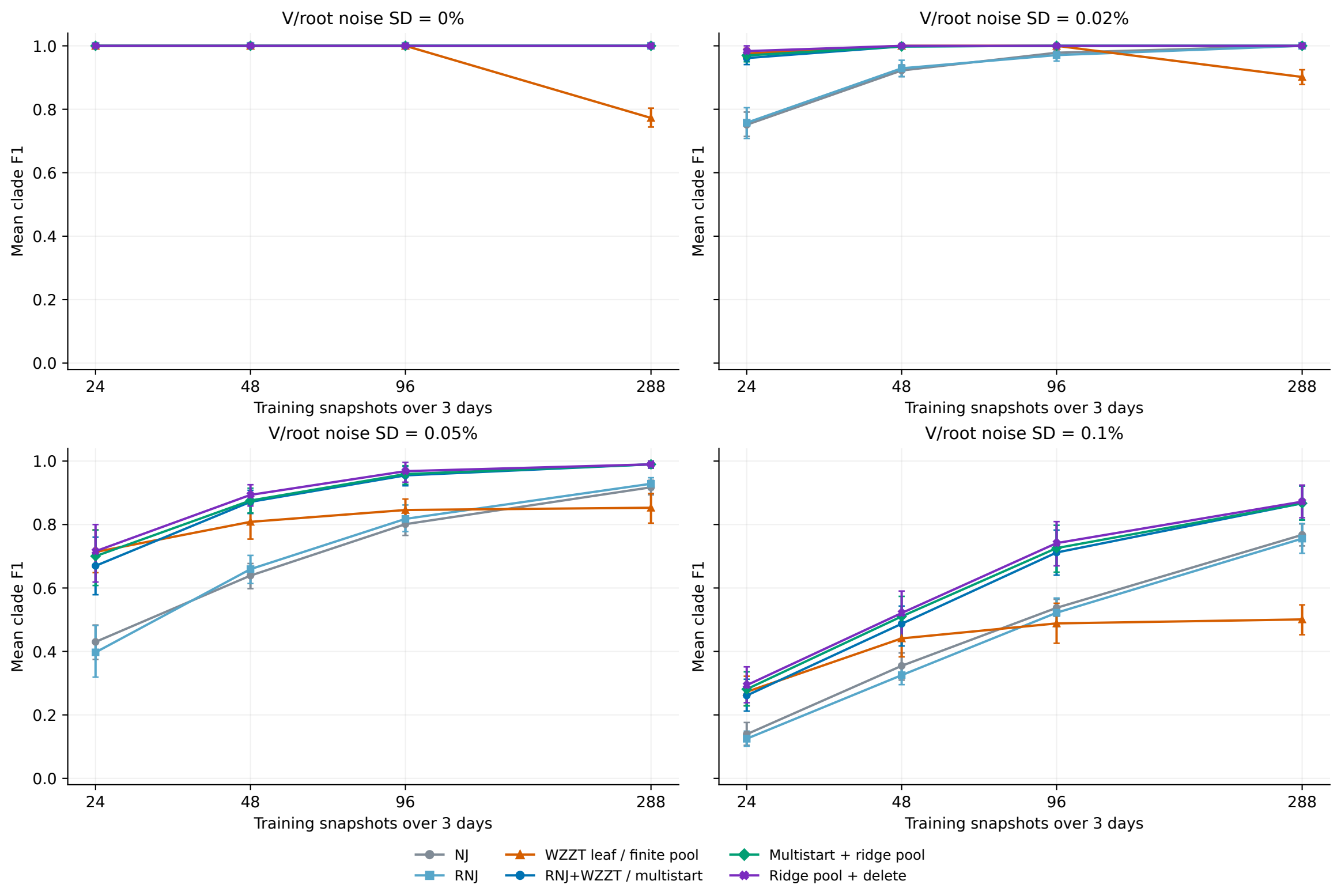
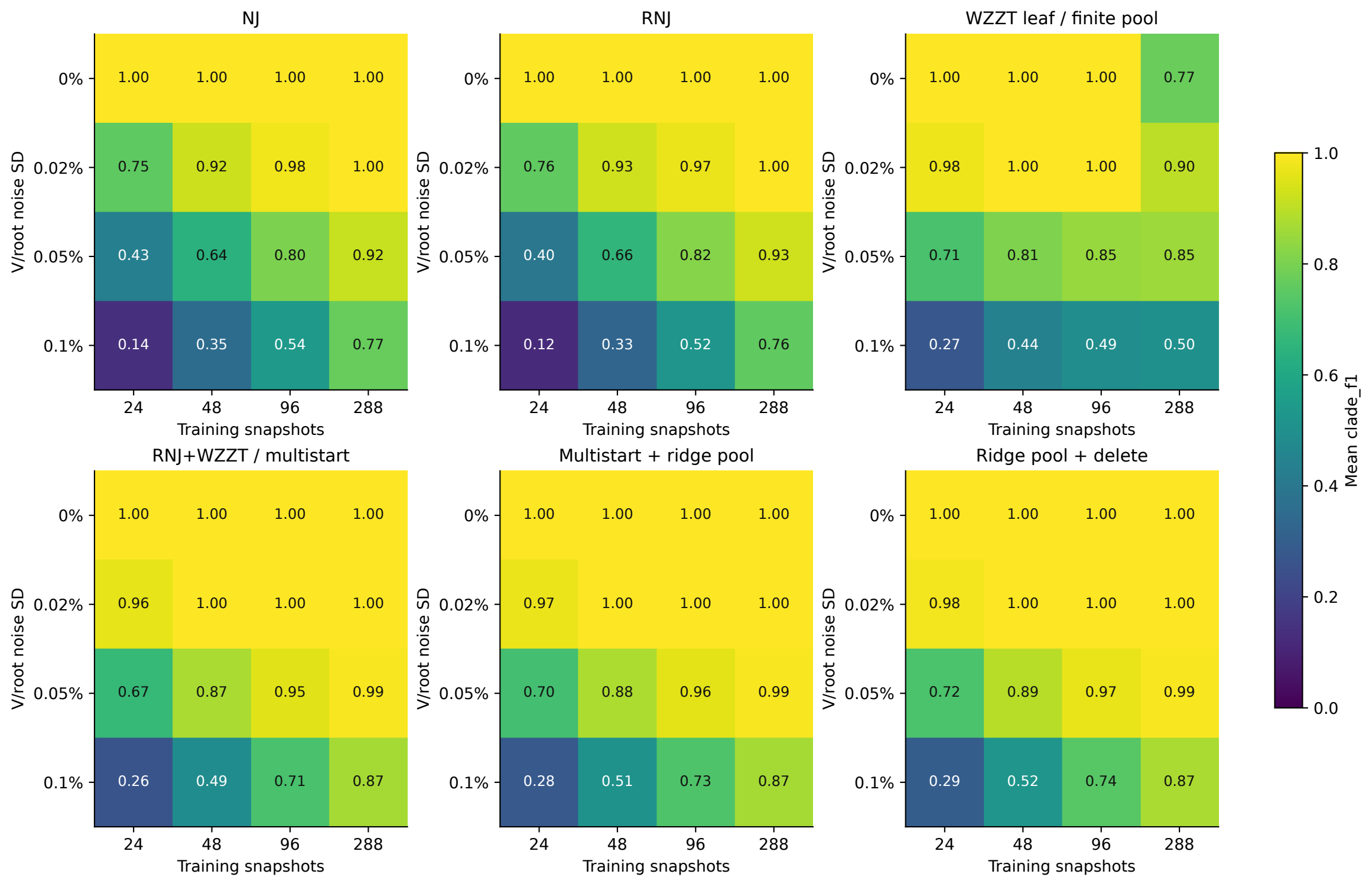
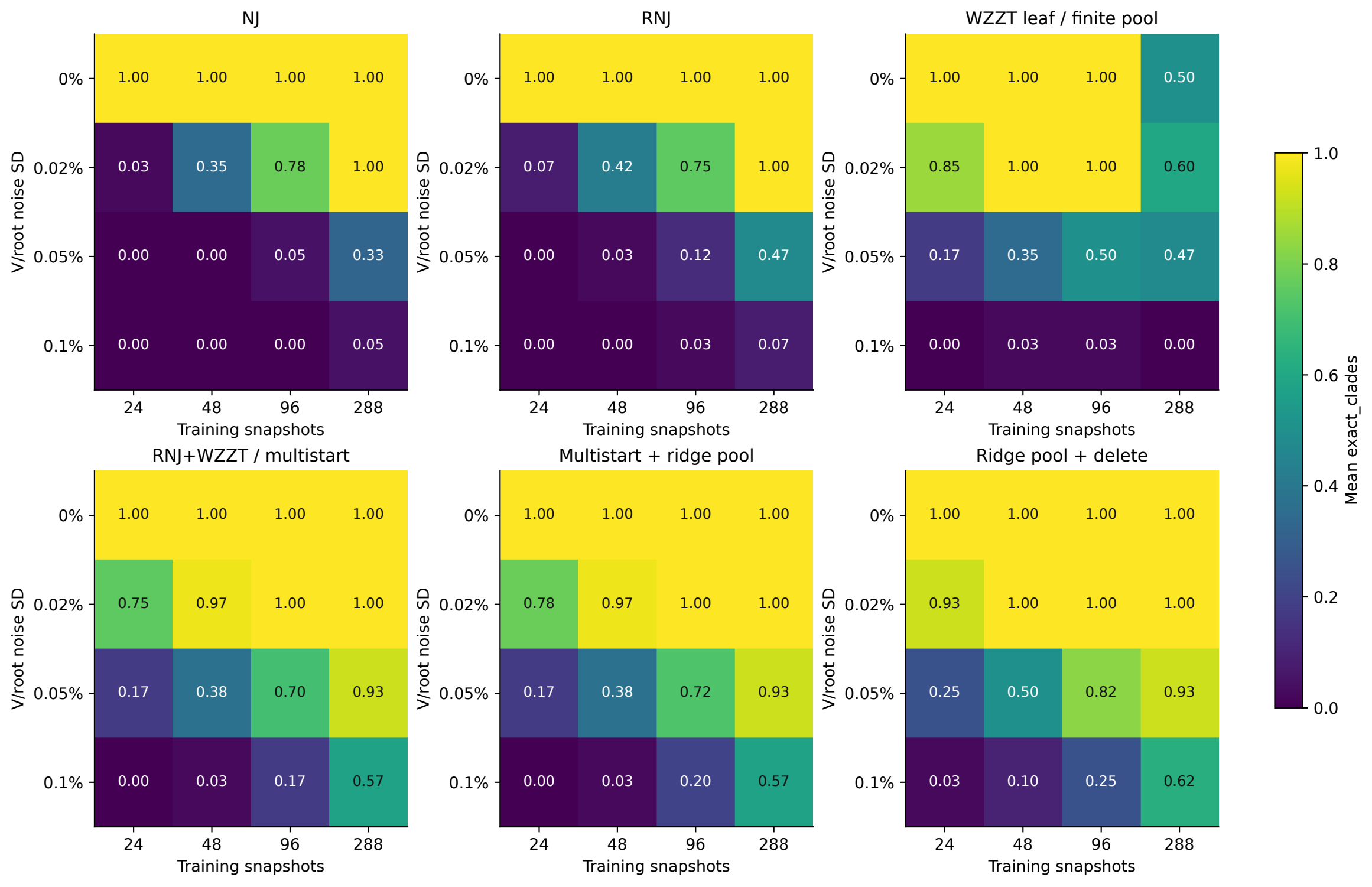
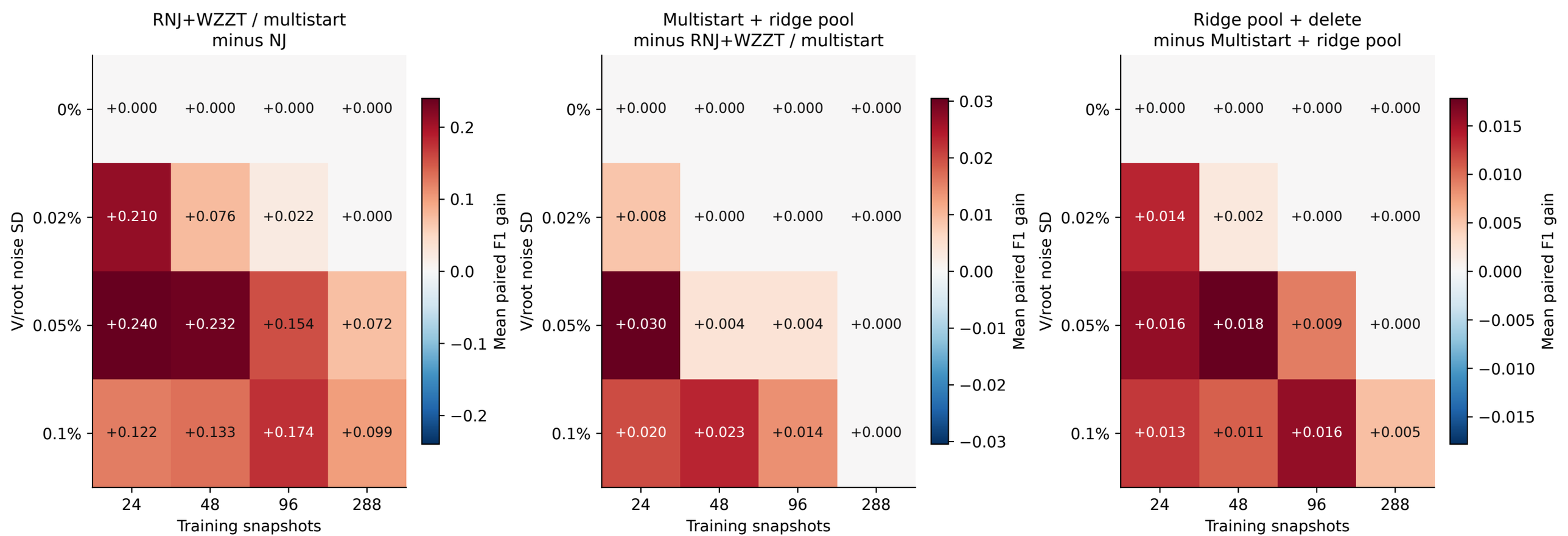

Main panel: 640 inputs per method; 10 independent data bundles
Mean clade F1
Exact nontrivial clade family
NJ
RNJ
RNJ bank (L1)
WZZT leaf / finite pool
WZZT leaf + RNJ bank
Single RNJ start
RNJ+WZZT / multistart
Multistart + pool restart
Multistart + ridge pool
Multistart + delete
Ridge pool + delete
0.765
0.762
0.826
0.786
0.816
0.837
0.861
0.861
0.867
0.867
0.874
0.411
0.436
0.608
0.531
0.598
0.616
0.667
0.667
0.672
0.709
0.714
0.0
0.2
0.4
0.6
0.8
1.0

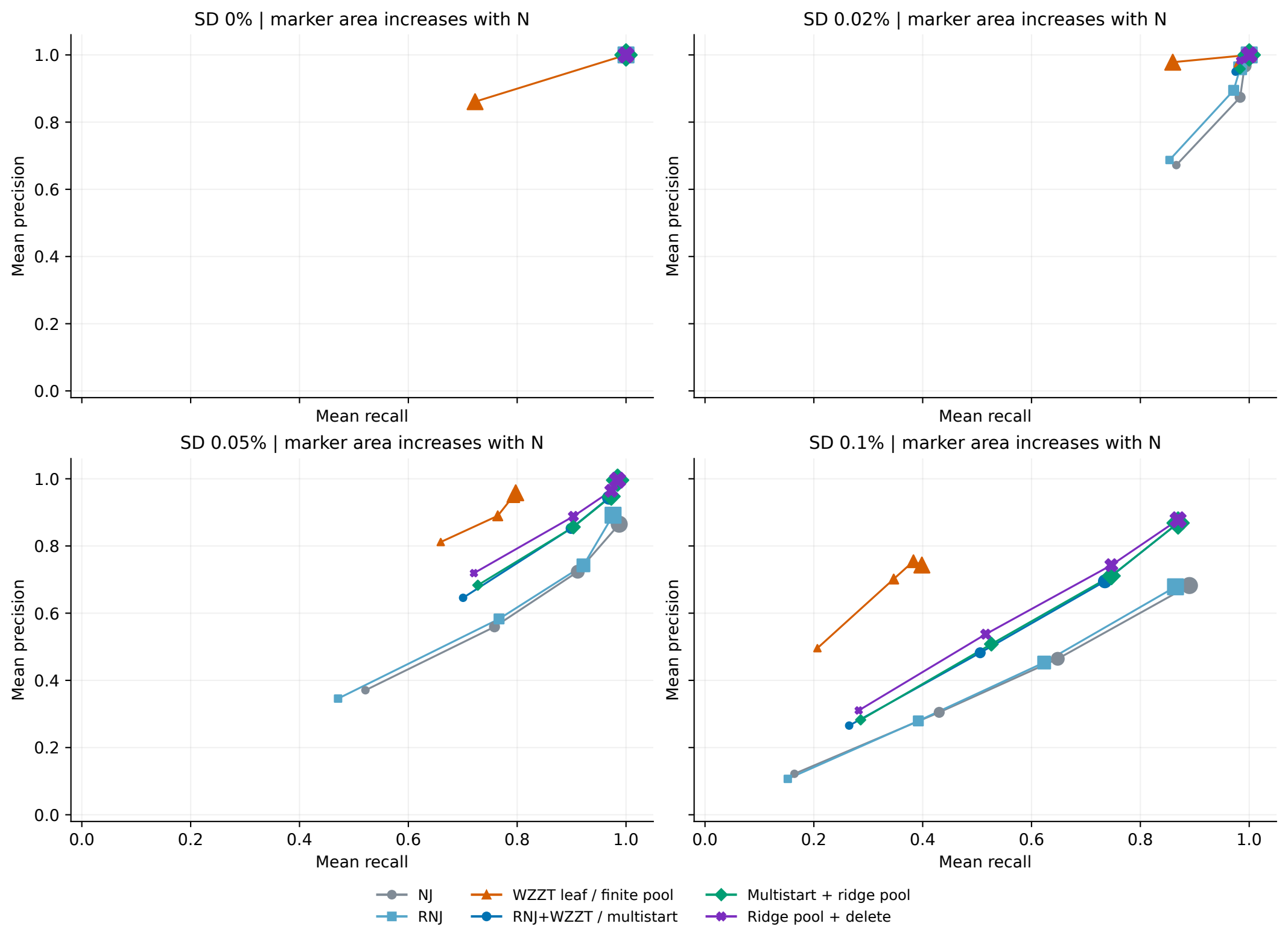

V/root noise SD = 0%
V/root noise SD = 0.02%
V/root noise SD = 0.05%
V/root noise SD = 0.1%
Mean clade F1
Training snapshots over 3 days
0.0
0.2
0.4
0.6
0.8
1.0
24
48
96
288
NJ
RNJ
WZZT leaf / finite pool
RNJ+WZZT / multistart
Multistart + ridge pool
Ridge pool + delete

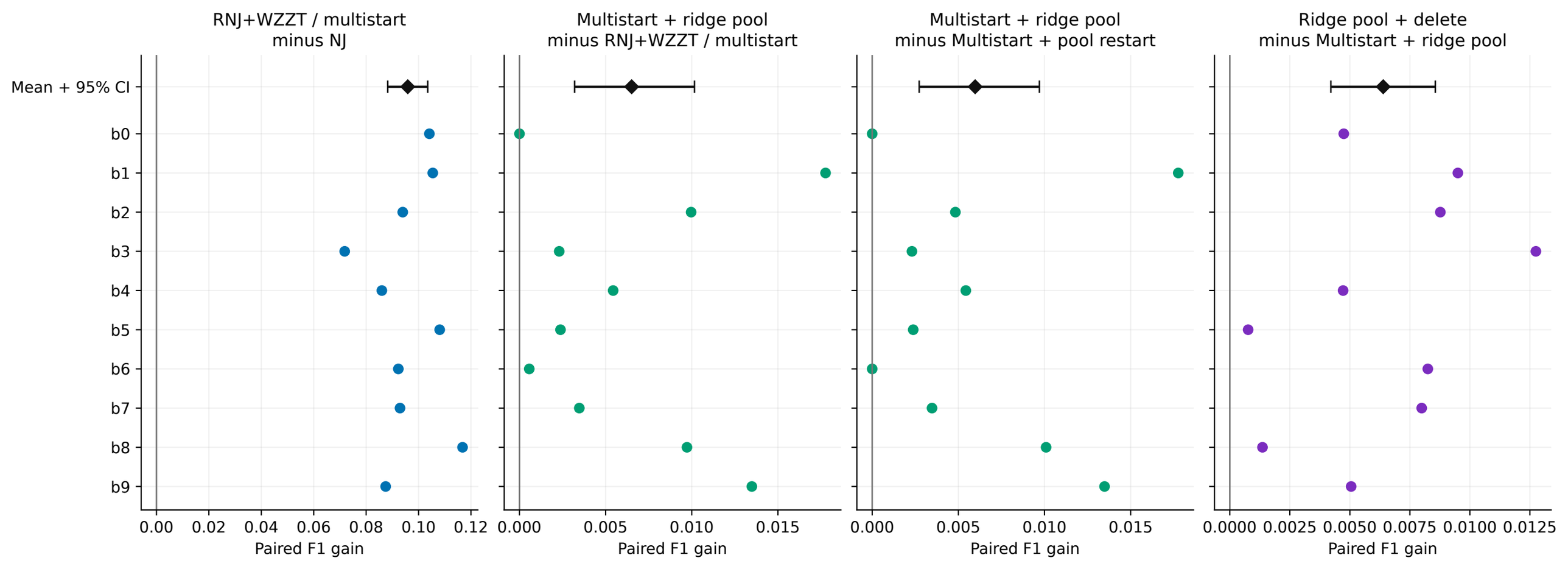
paper15 | SD 0%
paper15 | SD 0.02%
paper15 | SD 0.05%
paper15 | SD 0.1%
soumalas11 | SD 0%
soumalas11 | SD 0.02%
soumalas11 | SD 0.05%
soumalas11 | SD 0.1%
flynn16 | SD 0%
flynn16 | SD 0.02%
flynn16 | SD 0.05%
flynn16 | SD 0.1%
pengwah18 | SD 0%
pengwah18 | SD 0.02%
pengwah18 | SD 0.05%
pengwah18 | SD 0.1%
Clade F1
Training snapshots over 3 days
NJ
RNJ+WZZT / multistart
Ridge pool + delete

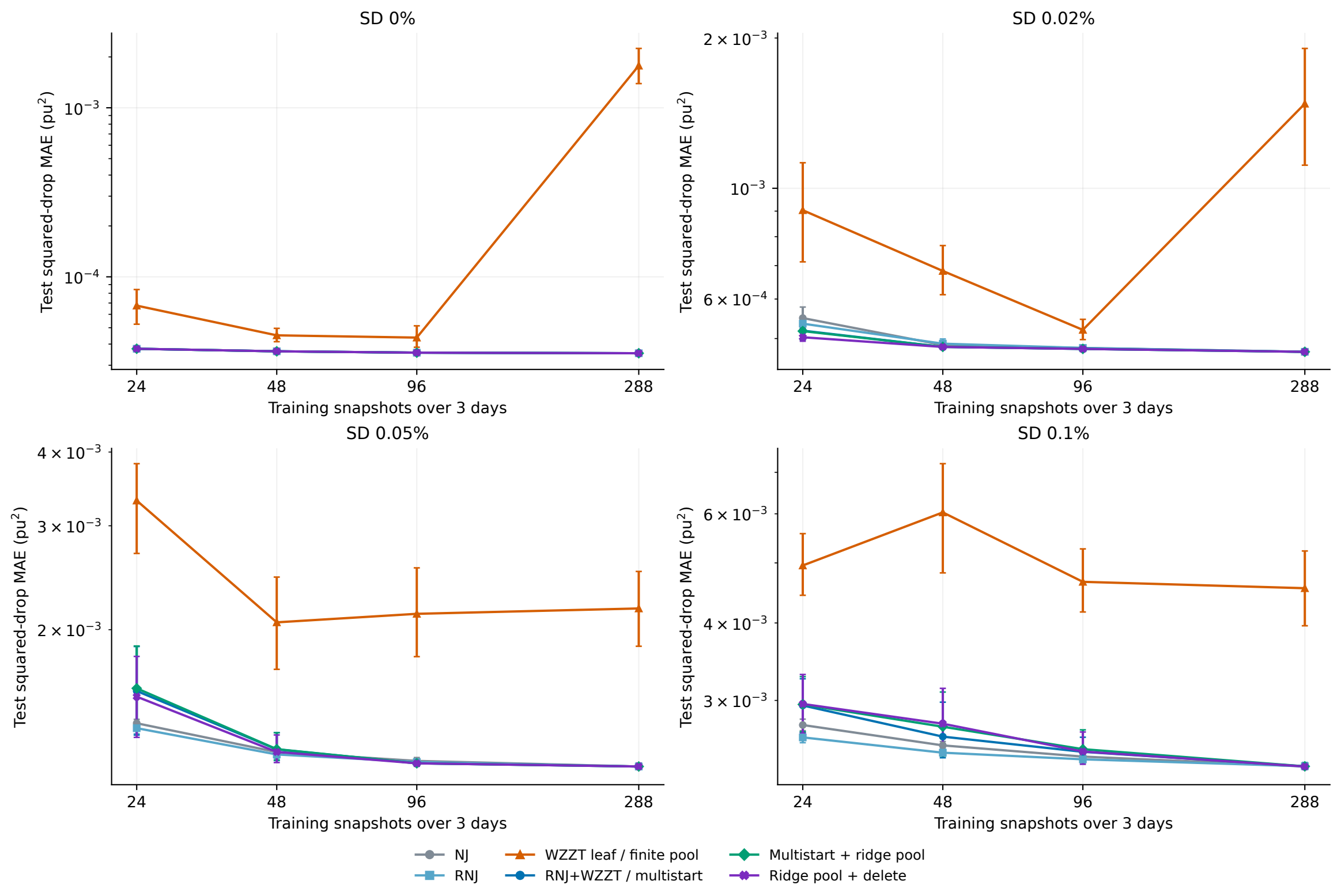

NJ
RNJ
WZZT leaf / finite pool
RNJ+WZZT / multistart
Multistart + ridge pool
Ridge pool + delete
V/root noise SD
0%
0.02%
0.05%
0.1%
Training snapshots
24
48
96
288
1.00 1.00 1.00 1.00
0.75 0.92 0.98 1.00
0.43 0.64 0.80 0.92
0.14 0.35 0.54 0.77
1.00 1.00 1.00 1.00
0.76 0.93 0.97 1.00
0.40 0.66 0.82 0.93
0.12 0.33 0.52 0.76
1.00 1.00 1.00 0.77
0.98 1.00 1.00 0.90
0.71 0.81 0.85 0.85
0.27 0.44 0.49 0.50
1.00 1.00 1.00 1.00
0.96 1.00 1.00 1.00
0.67 0.87 0.95 0.99
0.26 0.49 0.71 0.87
1.00 1.00 1.00 1.00
0.97 1.00 1.00 1.00
0.70 0.88 0.96 0.99
0.28 0.51 0.73 0.87
1.00 1.00 1.00 1.00
0.98 1.00 1.00 1.00
0.72 0.89 0.97 0.99
0.29 0.52 0.74 0.87
Mean clade_f1
1.0
0.8
0.6
0.4
0.2
0.0

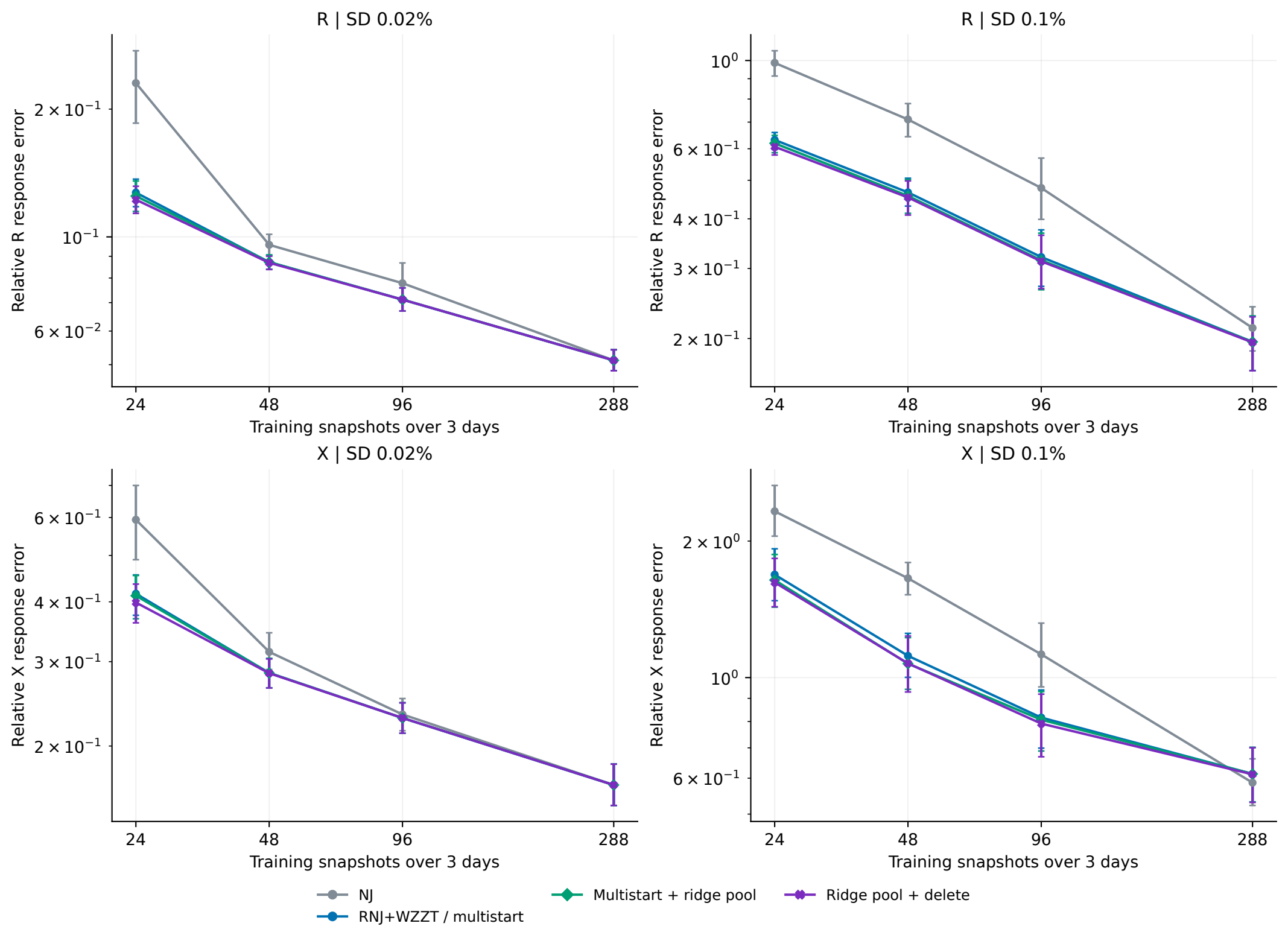

NJ
RNJ
WZZT leaf / finite pool
RNJ+WZZT / multistart
Multistart + ridge pool
Ridge pool + delete
V/root noise SD
0%
0.02%
0.05%
0.1%
Training snapshots
24
48
96
288
1.00 1.00 1.00 1.00
0.03 0.35 0.78 1.00
0.00 0.00 0.05 0.33
0.00 0.00 0.00 0.05
1.00 1.00 1.00 1.00
0.07 0.42 0.75 1.00
0.00 0.03 0.12 0.47
0.00 0.00 0.03 0.07
1.00 1.00 1.00 0.50
0.85 1.00 1.00 0.60
0.17 0.35 0.50 0.47
0.00 0.03 0.03 0.00
1.00 1.00 1.00 1.00
0.75 0.97 1.00 1.00
0.17 0.38 0.70 0.93
0.00 0.03 0.17 0.57
1.00 1.00 1.00 1.00
0.78 0.97 1.00 1.00
0.17 0.38 0.72 0.93
0.00 0.03 0.20 0.57
1.00 1.00 1.00 1.00
0.93 1.00 1.00 1.00
0.25 0.50 0.82 0.93
0.03 0.10 0.25 0.62
Mean exact_clades
1.0
0.8
0.6
0.4
0.2
0.0

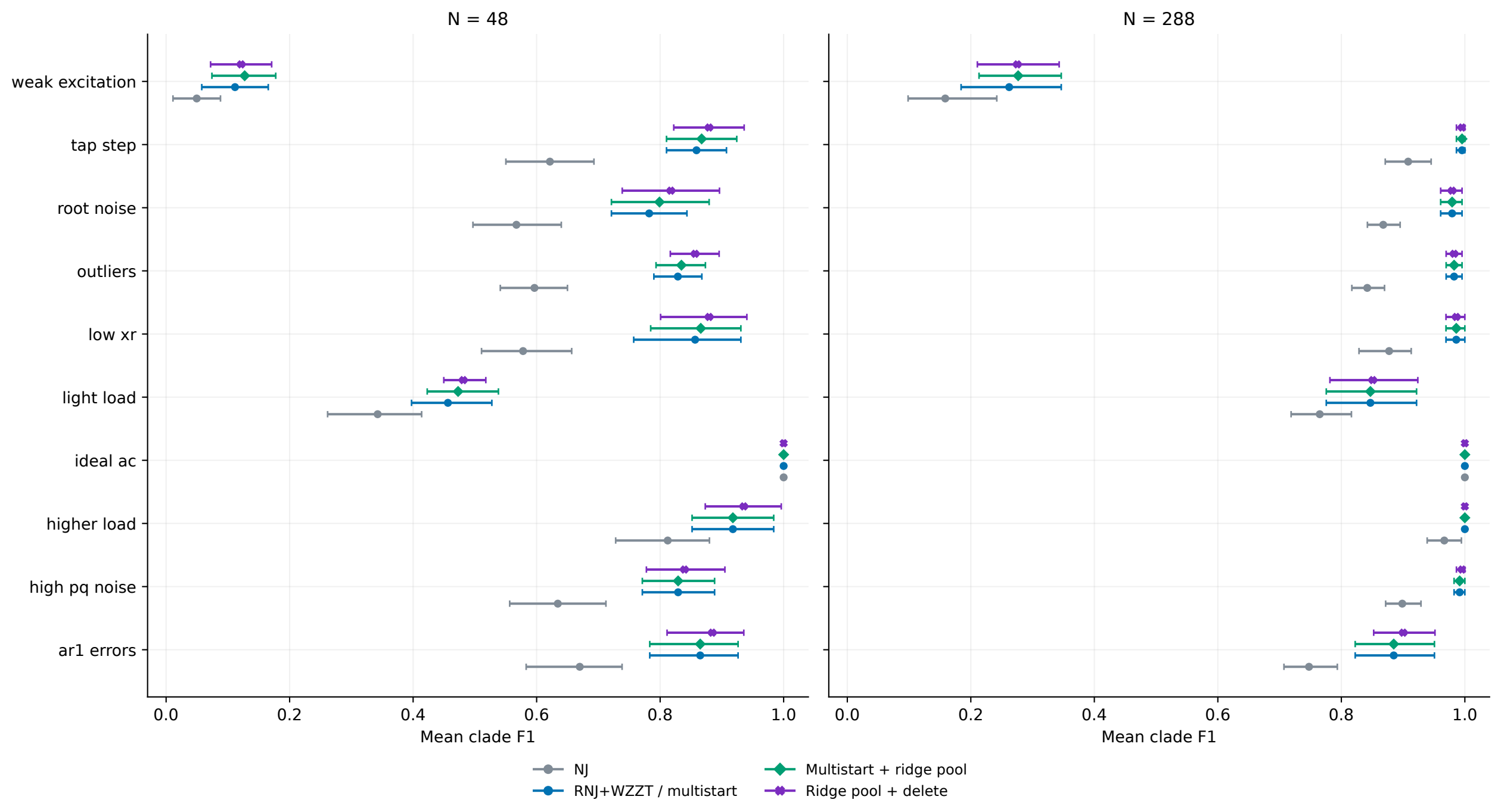

RNJ+WZZT / multistart
minus NJ
Multistart + ridge pool
minus RNJ+WZZT / multistart
Ridge pool + delete
minus Multistart + ridge pool
V/root noise SD
0%
0.02%
0.05%
0.1%
Training snapshots
24
48
96
288
Mean paired F1 gain
+0.000 +0.000 +0.000 +0.000
+0.210 +0.076 +0.022 +0.000
+0.240 +0.232 +0.154 +0.072
+0.122 +0.133 +0.174 +0.099
0.2
0.1
0.0
−0.1
−0.2
+0.000 +0.000 +0.000 +0.000
+0.008 +0.000 +0.000 +0.000
+0.030 +0.004 +0.004 +0.000
+0.020 +0.023 +0.014 +0.000
0.03
0.02
0.01
0.00
−0.01
−0.02
−0.03
+0.000 +0.000 +0.000 +0.000
+0.014 +0.002 +0.000 +0.000
+0.016 +0.018 +0.009 +0.000
+0.013 +0.011 +0.016 +0.005
0.015
0.010
0.005
0.000
−0.005
−0.010
−0.015

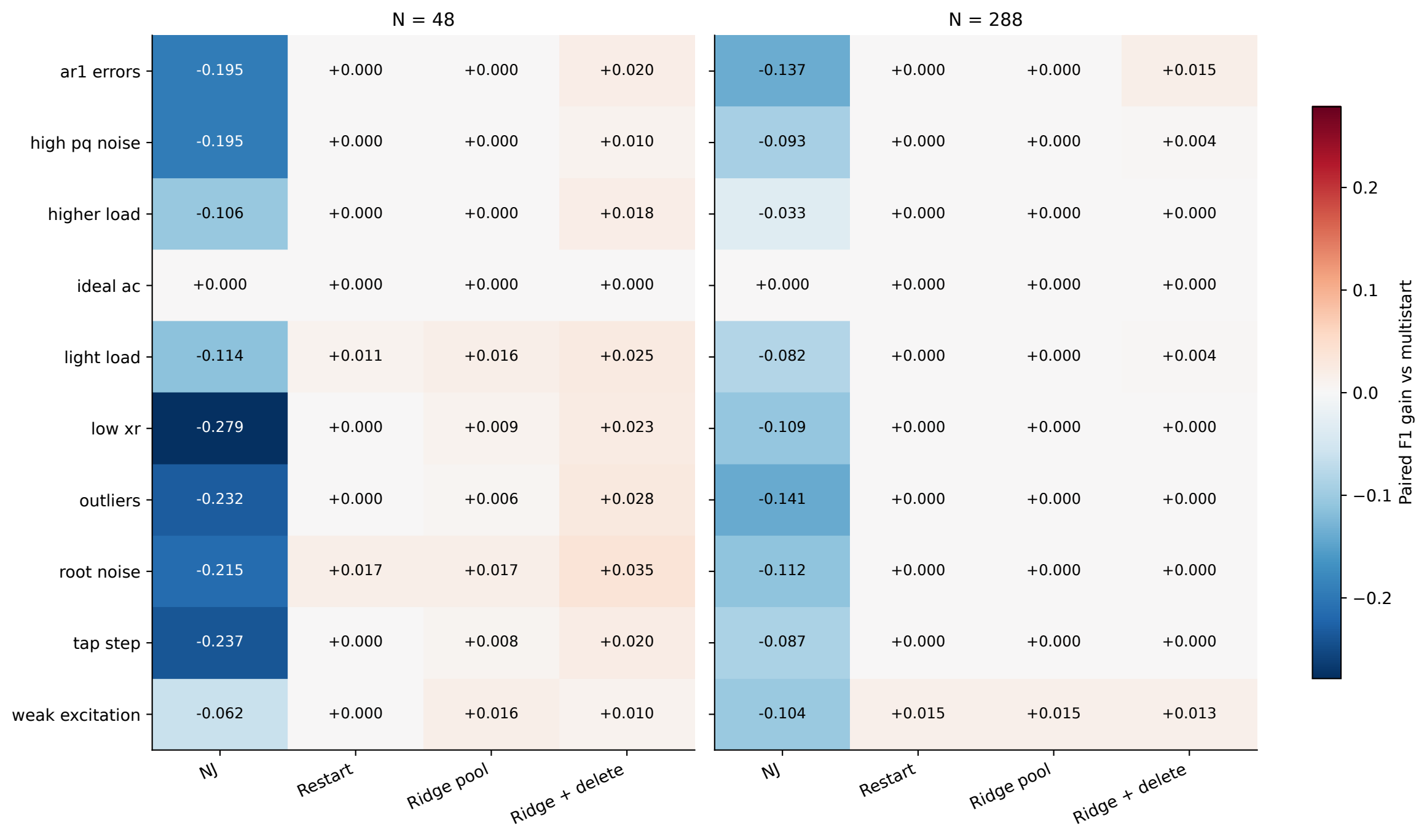

SD 0% | marker area increases with N
SD 0.02% | marker area increases with N
SD 0.05% | marker area increases with N
SD 0.1% | marker area increases with N
Mean precision
Mean recall
0.0
0.2
0.4
0.6
0.8
1.0
NJ
RNJ
WZZT leaf / finite pool
RNJ+WZZT / multistart
Multistart + ridge pool
Ridge pool + delete

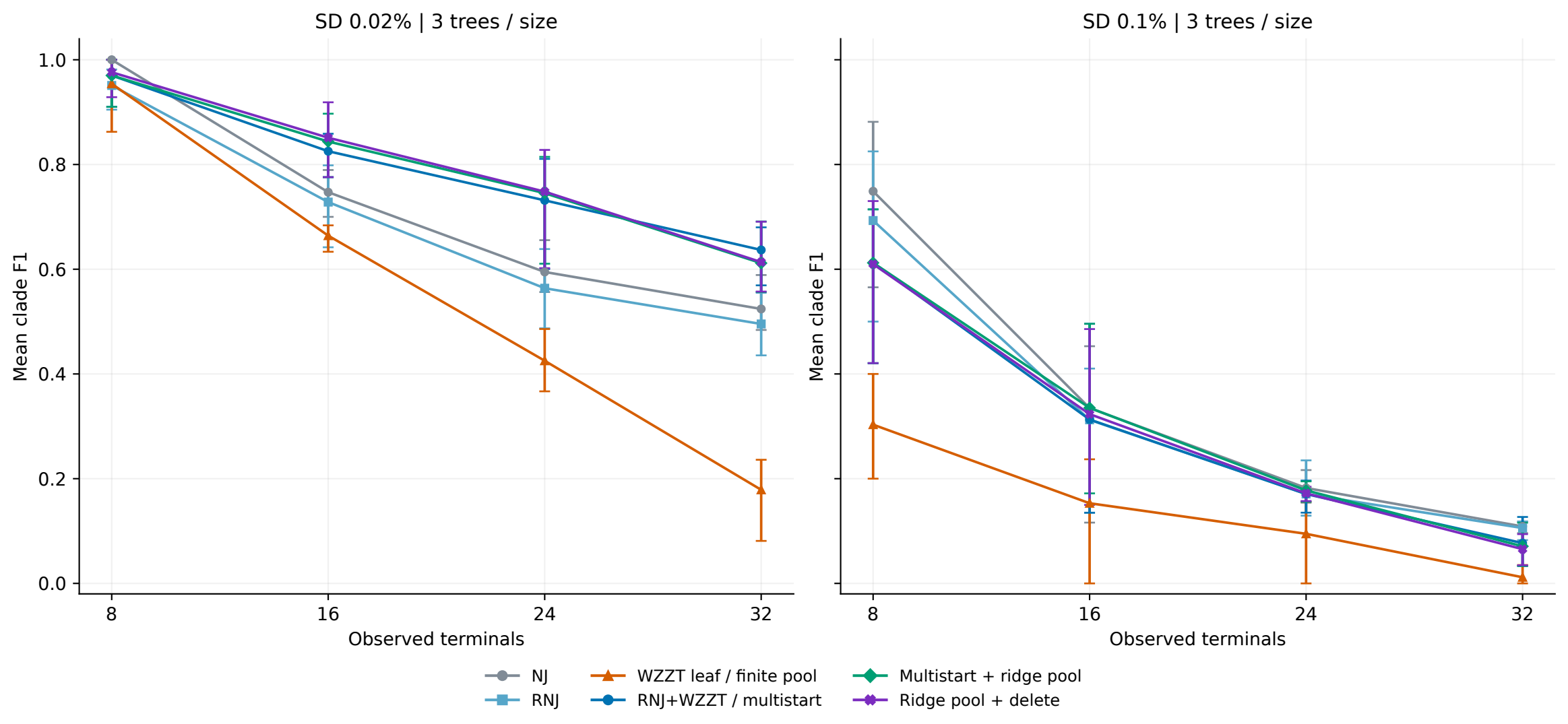

RNJ+WZZT / multistart
minus NJ
Multistart + ridge pool
minus RNJ+WZZT / multistart
Multistart + ridge pool
minus Multistart + pool restart
Ridge pool + delete
minus Multistart + ridge pool
Mean + 95% CI
b0
b1
b2
b3
b4
b5
b6
b7
b8
b9
0.00 0.02 0.04 0.06 0.08 0.10 0.12
0.000 0.005 0.010 0.015
0.000 0.005 0.010 0.015
0.0000 0.0025 0.0050 0.0075 0.0100 0.0125
Paired F1 gain
Paired F1 gain
Paired F1 gain
Paired F1 gain

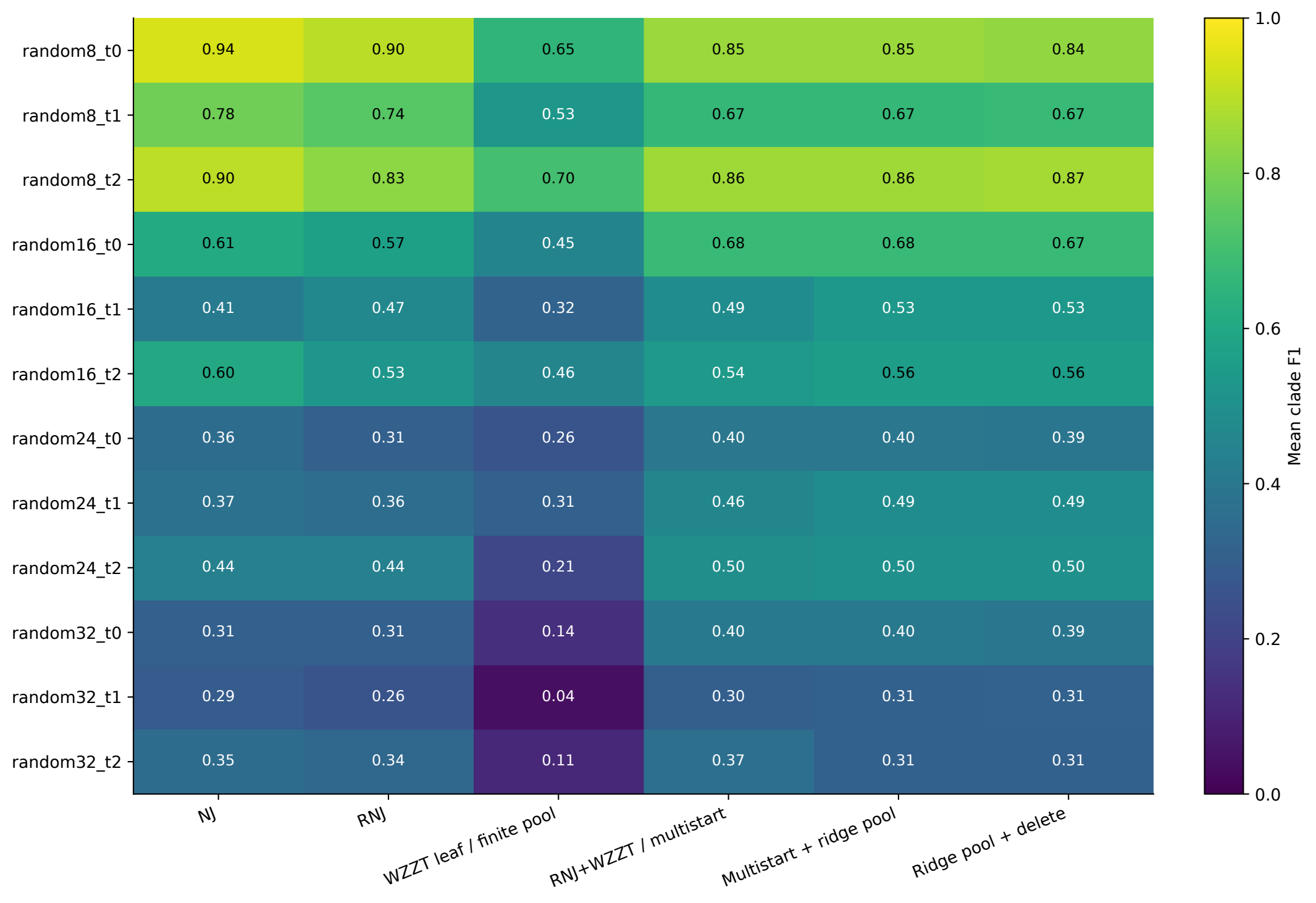

SD 0%
SD 0.02%
SD 0.05%
SD 0.1%
Test squared-drop MAE (pu²)
Training snapshots over 3 days
24
48
96
288
NJ
RNJ
WZZT leaf / finite pool
RNJ+WZZT / multistart
Multistart + ridge pool
Ridge pool + delete

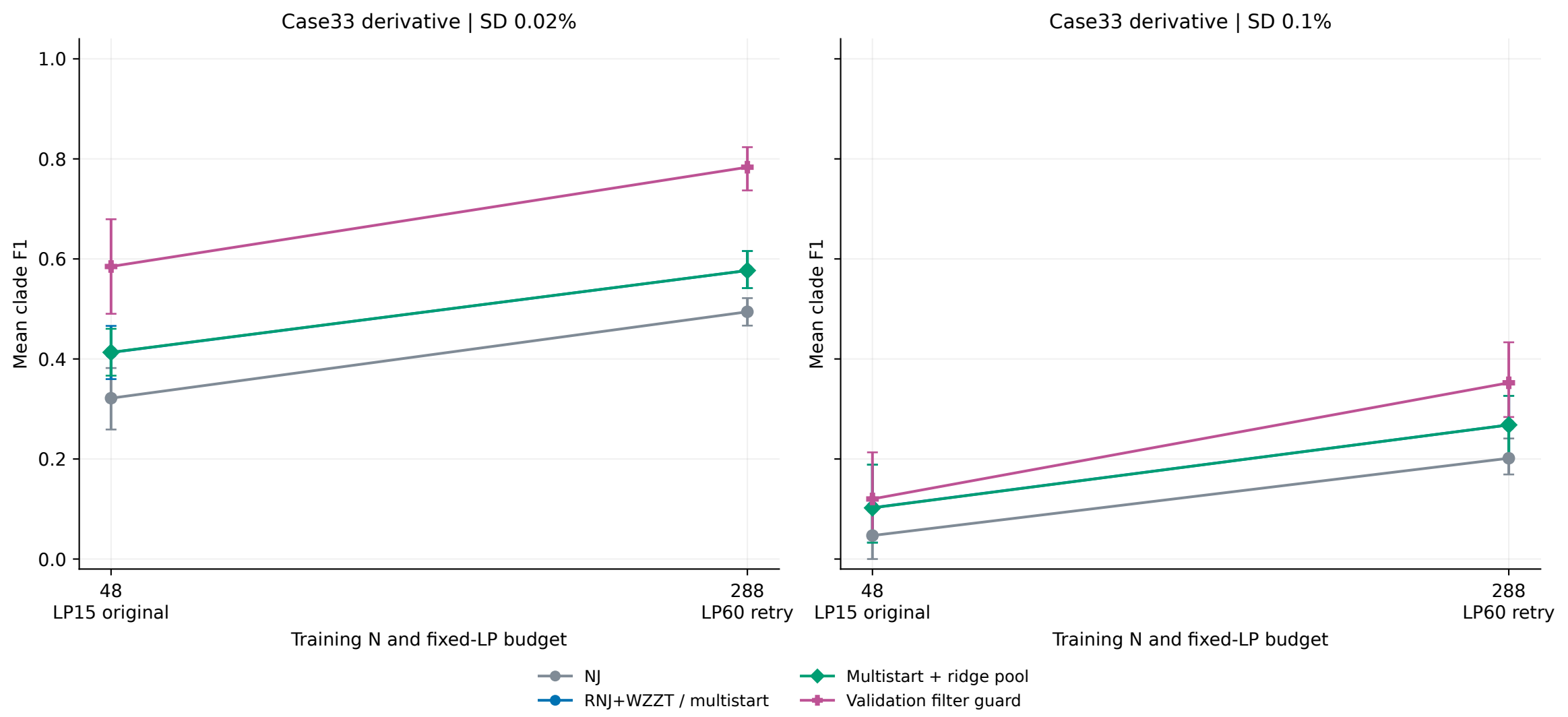

R | SD 0.02%
R | SD 0.1%
X | SD 0.02%
X | SD 0.1%
Relative R response error
Relative X response error
Training snapshots over 3 days
24
48
96
288
NJ
RNJ+WZZT / multistart
Multistart + ridge pool
Ridge pool + delete

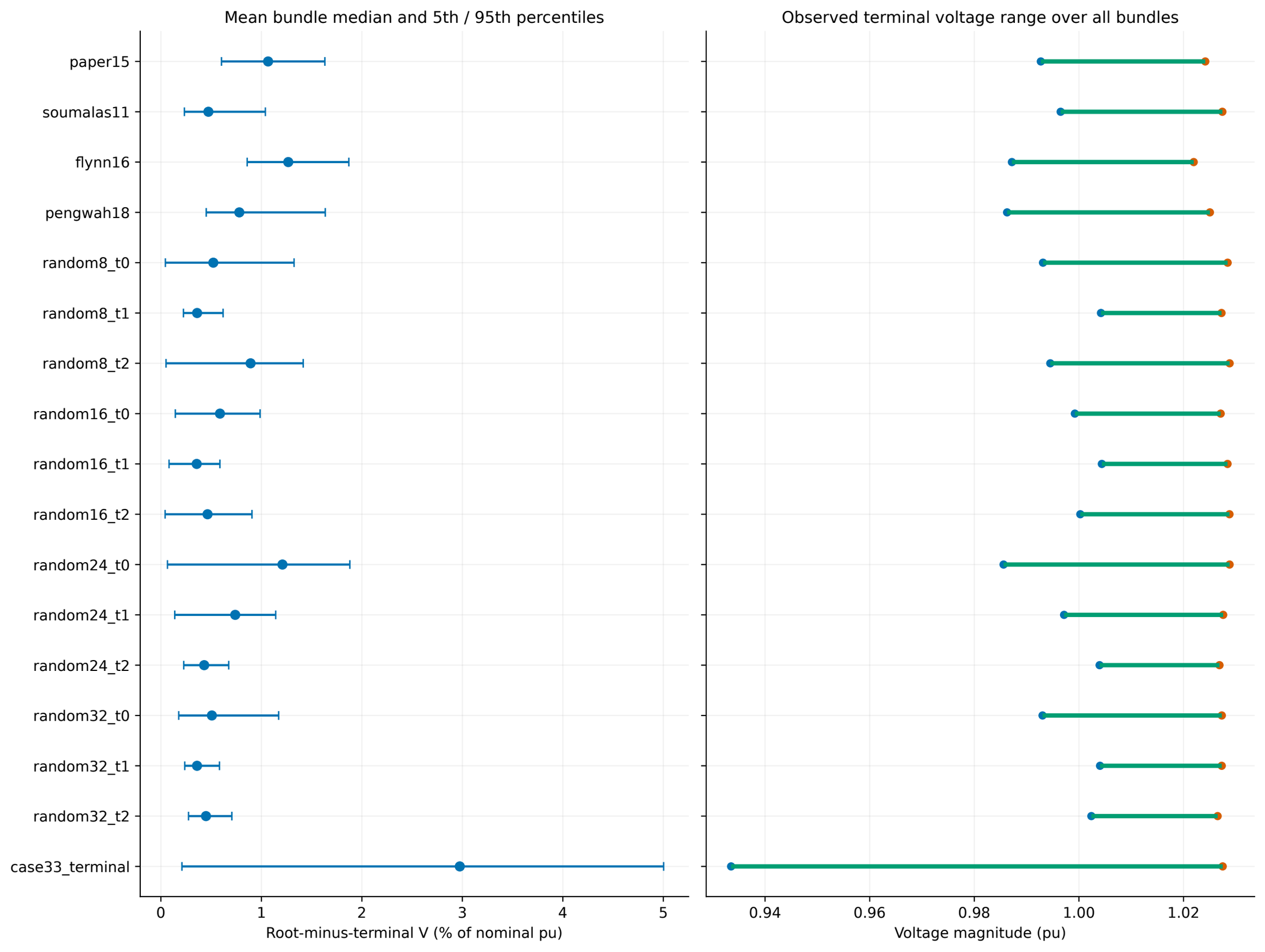

N = 48
N = 288
weak excitation
tap step
root noise
outliers
low xr
light load
ideal ac
higher load
high pq noise
ar1 errors
0.0
0.2
0.4
0.6
0.8
1.0
Mean clade F1
NJ
RNJ+WZZT / multistart
Multistart + ridge pool
Ridge pool + delete

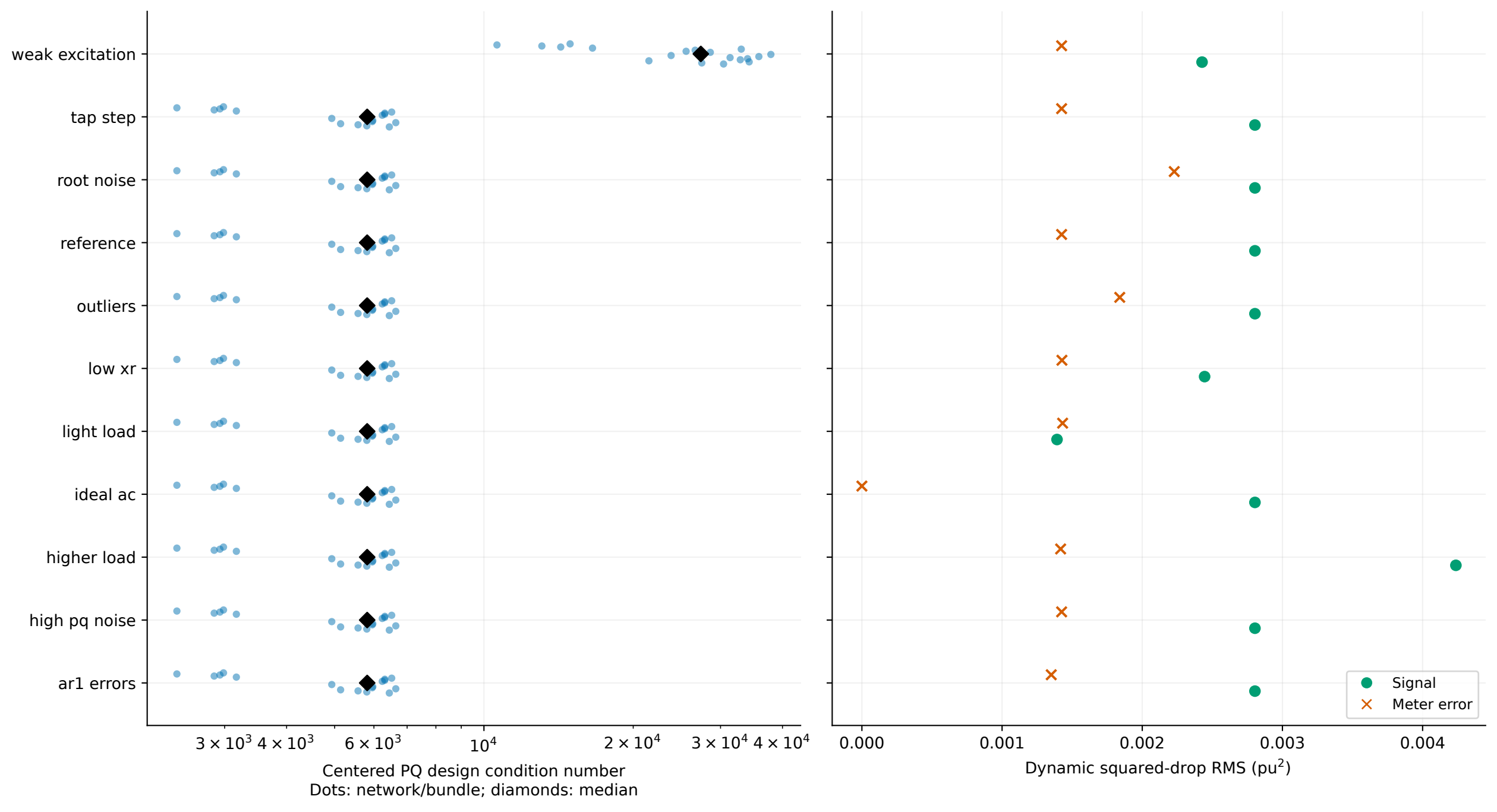

N = 48
N = 288
ar1 errors
high pq noise
higher load
ideal ac
light load
low xr
outliers
root noise
tap step
weak excitation
-0.195 +0.000 +0.000 +0.020
-0.195 +0.000 +0.000 +0.010
-0.106 +0.000 +0.000 +0.018
+0.000 +0.000 +0.000 +0.000
-0.114 +0.011 +0.016 +0.025
-0.279 +0.000 +0.009 +0.023
-0.232 +0.000 +0.006 +0.028
-0.215 +0.017 +0.017 +0.035
-0.237 +0.000 +0.008 +0.020
-0.062 +0.000 +0.016 +0.010
-0.137 +0.000 +0.000 +0.015
-0.093 +0.000 +0.000 +0.004
-0.033 +0.000 +0.000 +0.000
+0.000 +0.000 +0.000 +0.000
-0.082 +0.000 +0.000 +0.004
-0.109 +0.000 +0.000 +0.000
-0.141 +0.000 +0.000 +0.000
-0.112 +0.000 +0.000 +0.000
-0.087 +0.000 +0.000 +0.000
-0.104 +0.015 +0.015 +0.013
NJ
Restart
Ridge pool
Ridge + delete
NJ
Restart
Ridge pool
Ridge + delete
Paired F1 gain vs multistart
0.2
0.1
0.0
−0.1
−0.2

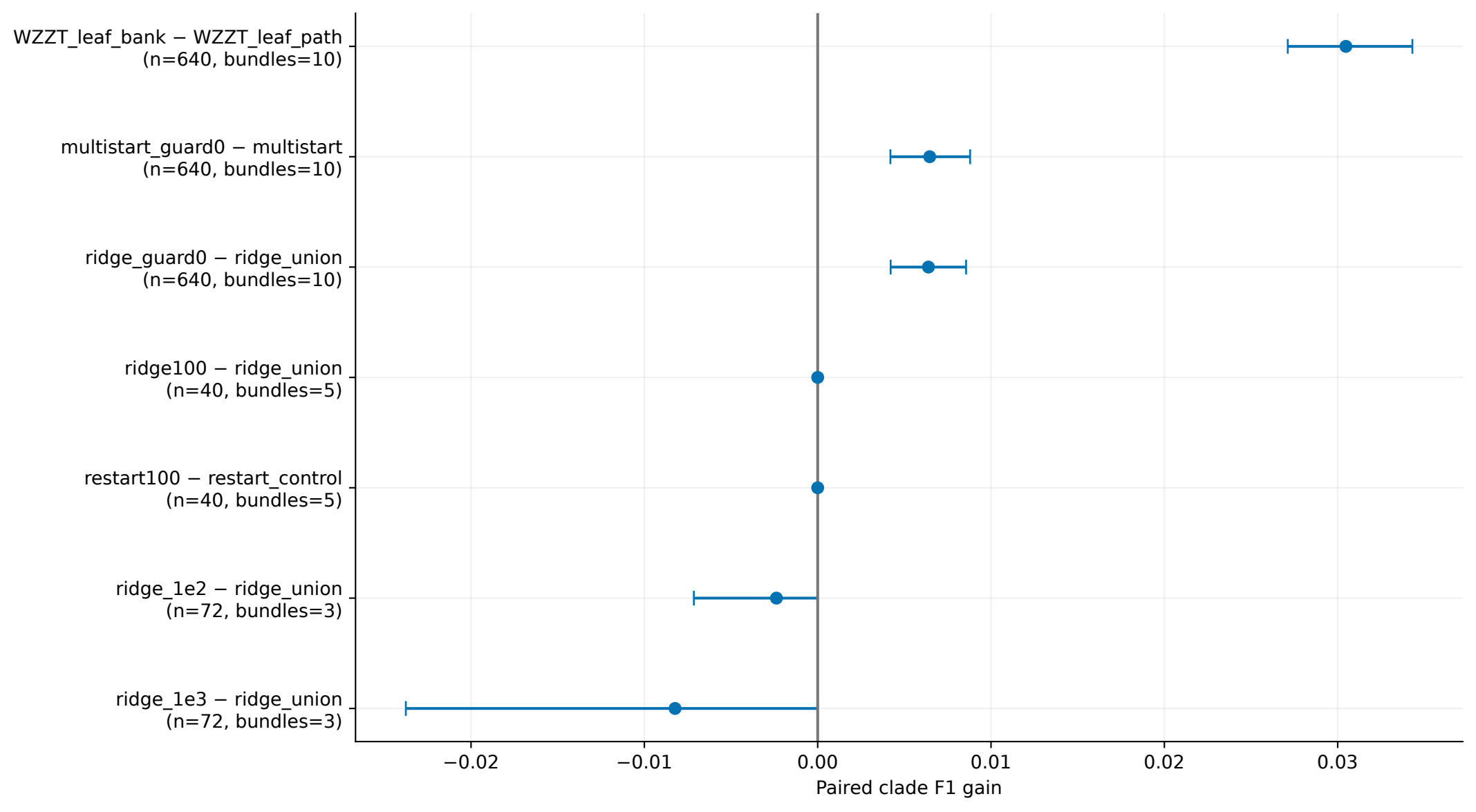

SD 0.02% | 3 trees / size
SD 0.1% | 3 trees / size
Mean clade F1
1.0
0.8
0.6
0.4
0.2
0.0
8
16
24
32
Observed terminals
NJ
RNJ
WZZT leaf / finite pool
RNJ+WZZT / multistart
Multistart + ridge pool
Ridge pool + delete

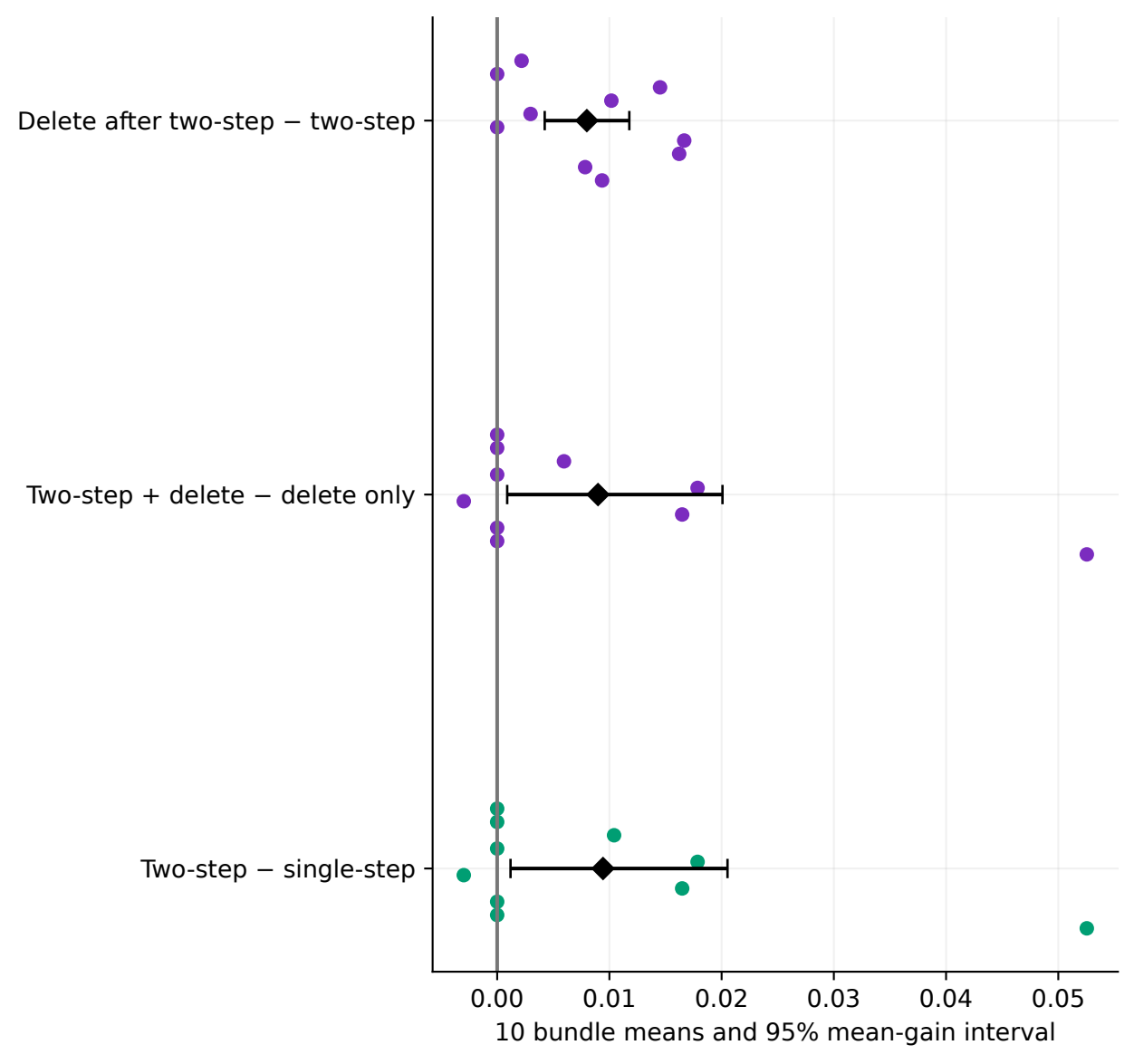

random8_t0
random8_t1
random8_t2
random16_t0
random16_t1
random16_t2
random24_t0
random24_t1
random24_t2
random32_t0
random32_t1
random32_t2
0.94 0.90 0.65 0.85 0.85 0.84
0.78 0.74 0.53 0.67 0.67 0.67
0.90 0.83 0.70 0.86 0.86 0.87
0.61 0.57 0.45 0.68 0.68 0.67
0.41 0.47 0.32 0.49 0.53 0.53
0.60 0.53 0.46 0.54 0.56 0.56
0.36 0.31 0.26 0.40 0.40 0.39
0.37 0.36 0.31 0.46 0.49 0.49
0.44 0.44 0.21 0.50 0.50 0.50
0.31 0.31 0.14 0.40 0.40 0.39
0.29 0.26 0.04 0.30 0.31 0.31
0.35 0.34 0.11 0.37 0.31 0.31
NJ
RNJ
WZZT leaf / finite pool
RNJ+WZZT / multistart
Multistart + ridge pool
Ridge pool + delete
Mean clade F1
1.0
0.8
0.6
0.4
0.2
0.0

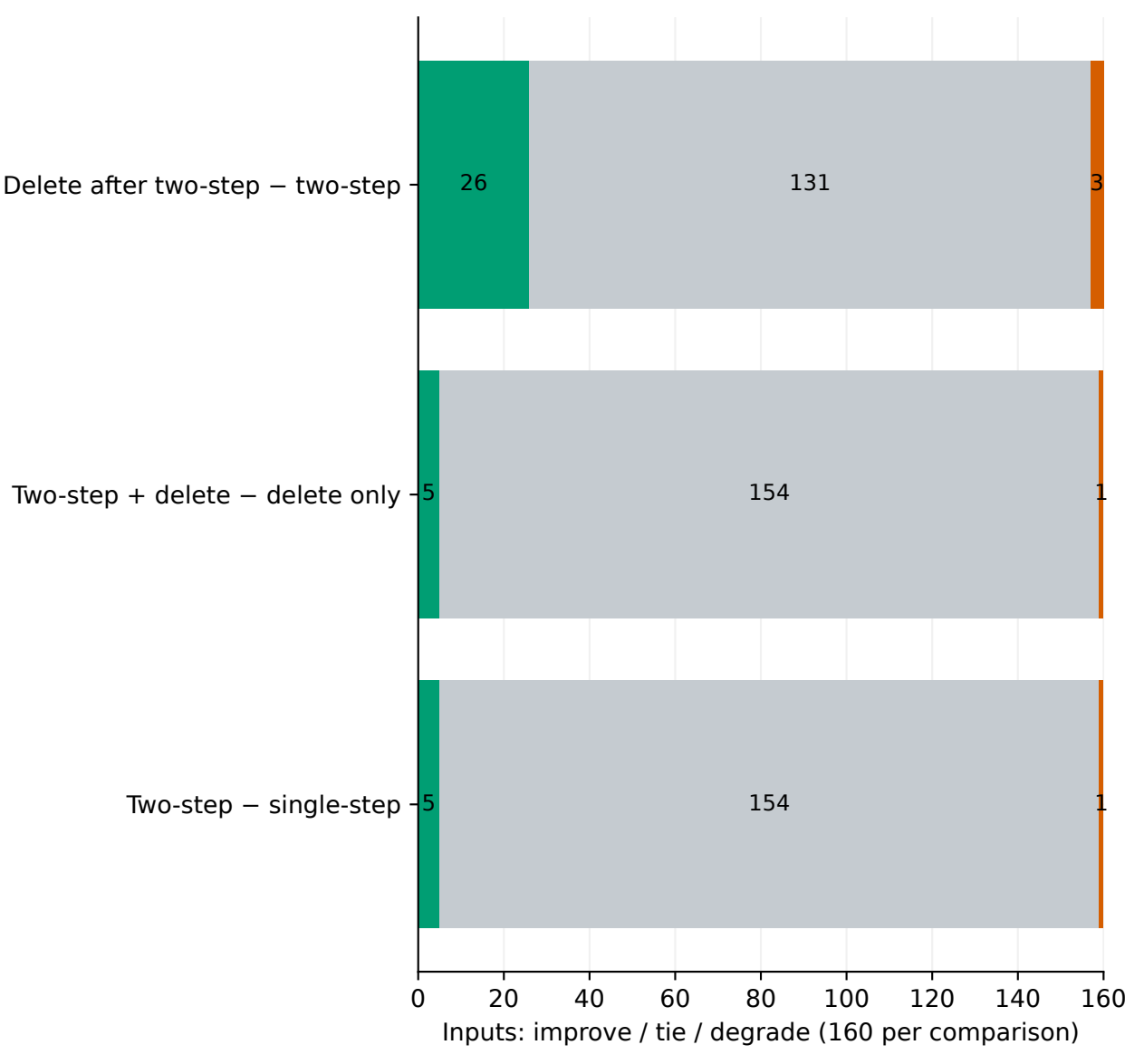

Case33 derivative | SD 0.02%
Case33 derivative | SD 0.1%
Mean clade F1
1.0
0.8
0.6
0.4
0.2
0.0
48
LP15 original
288
LP60 retry
Training N and fixed-LP budget
NJ
RNJ+WZZT / multistart
Multistart + ridge pool
Validation filter guard

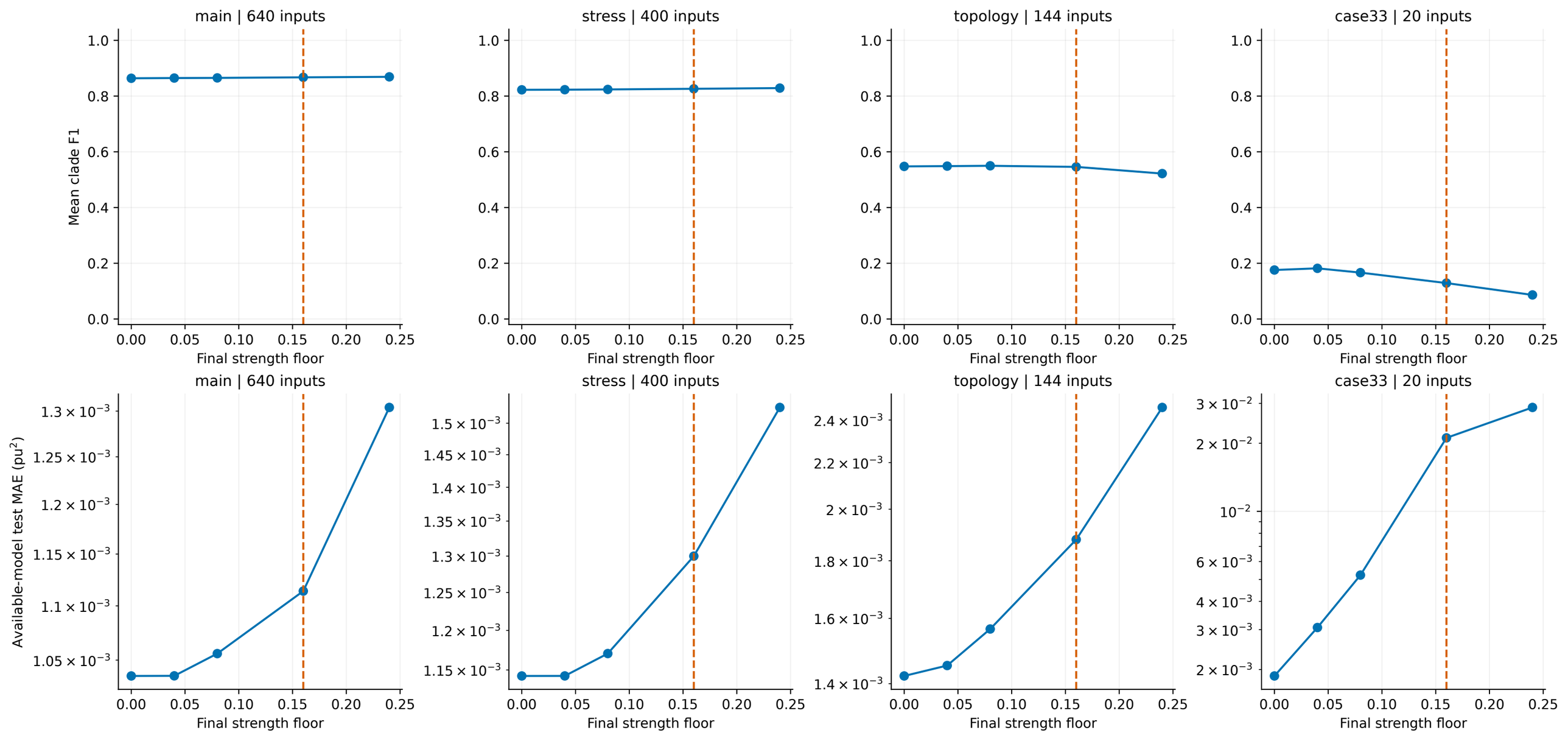

Mean bundle median and 5th / 95th percentiles
paper15
soumalas11
flynn16
pengwah18
random8_t0
random8_t1
random8_t2
random16_t0
random16_t1
random16_t2
random24_t0
random24_t1
random24_t2
random32_t0
random32_t1
random32_t2
case33_terminal
0
1
2
3
4
5
Root-minus-terminal V (% of nominal pu)
Observed terminal voltage range over all bundles
0.94
0.96
0.98
1.00
1.02
Voltage magnitude (pu)

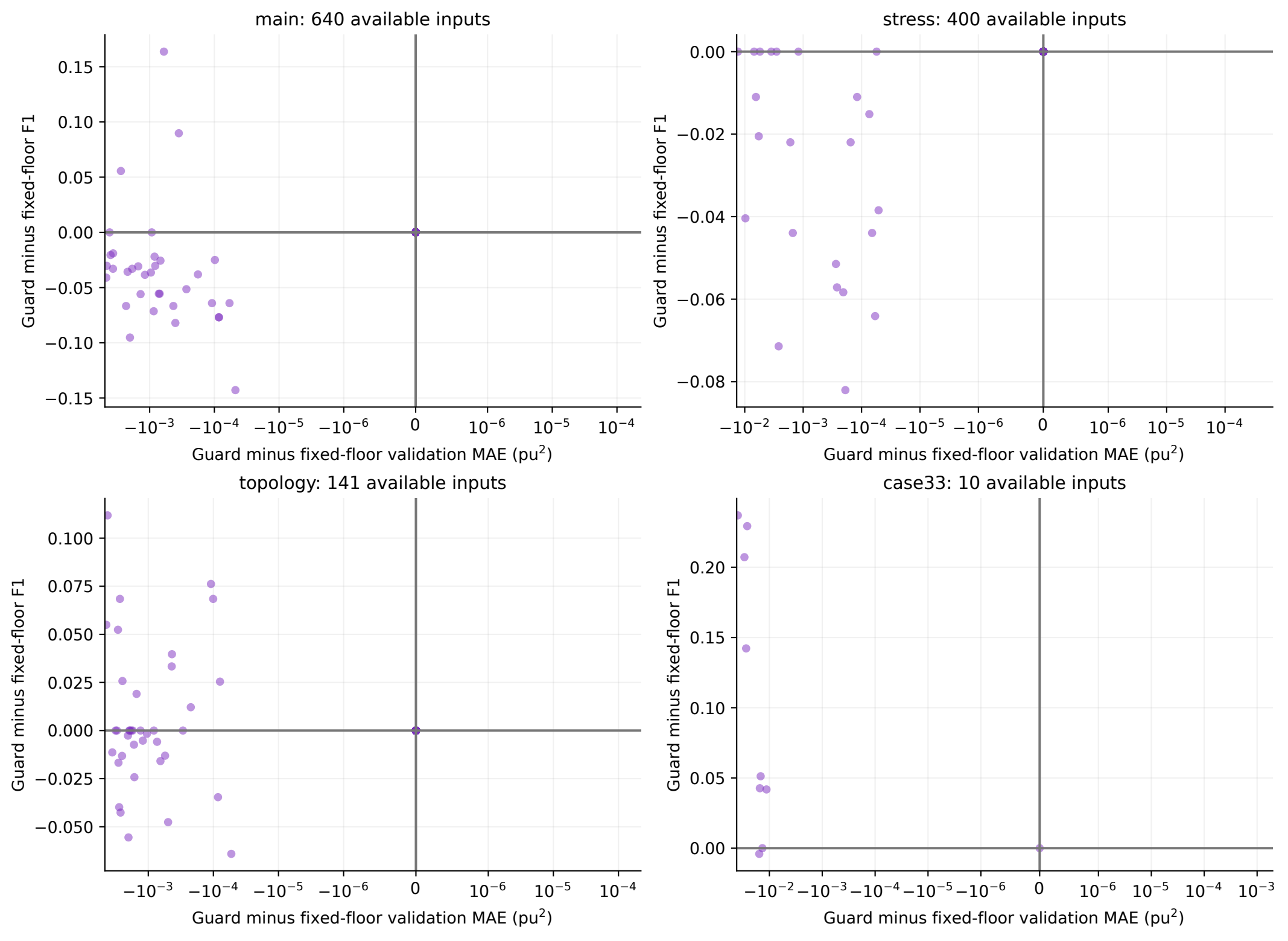

weak excitation
tap step
root noise
reference
outliers
low xr
light load
ideal ac
higher load
high pq noise
ar1 errors
$3 \times 10^3$ $4 \times 10^3$ $6 \times 10^3$ $10^4$ $2 \times 10^4$ $3 \times 10^4$ $4 \times 10^4$
Centered PQ design condition number
Dots: network/bundle; diamonds: median
0.000 0.001 0.002 0.003 0.004
Dynamic squared-drop RMS (pu$^2$)
Signal
Meter error

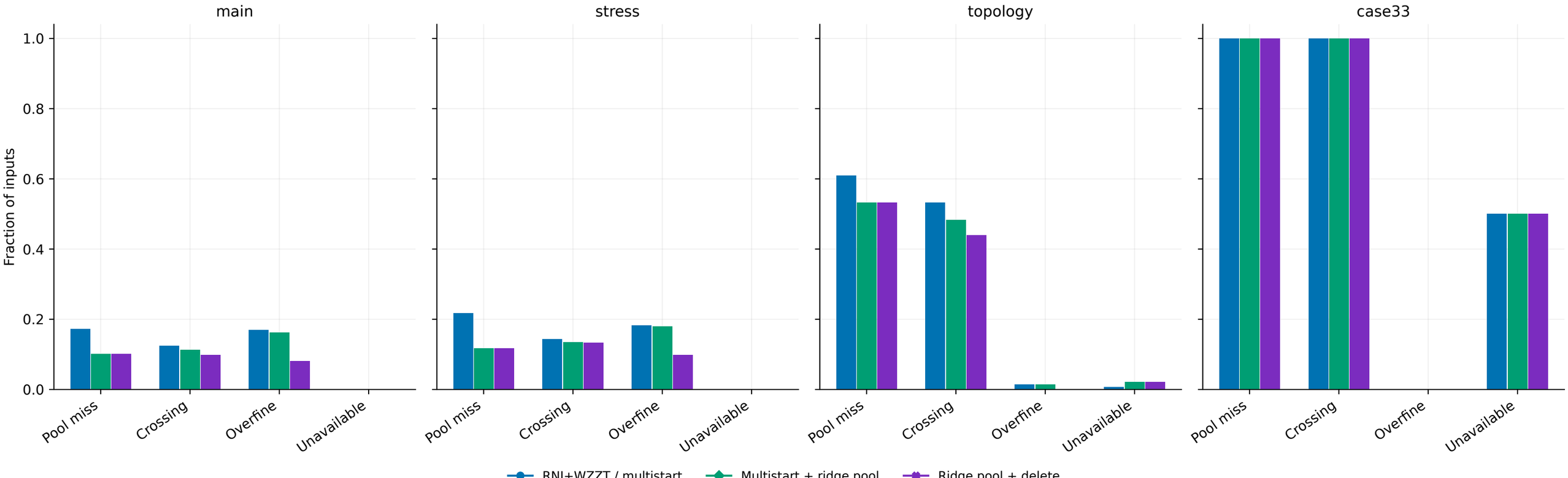

WZZT_leaf_bank − WZZT_leaf_path
(n=640, bundles=10)
multistart_guard0 − multistart
(n=640, bundles=10)
ridge_guard0 − ridge_union
(n=640, bundles=10)
ridge100 − ridge_union
(n=40, bundles=5)
restart100 − restart_control
(n=40, bundles=5)
ridge_1e2 − ridge_union
(n=72, bundles=3)
ridge_1e3 − ridge_union
(n=72, bundles=3)
−0.02
−0.01
0.00
0.01
0.02
0.03
Paired clade F1 gain

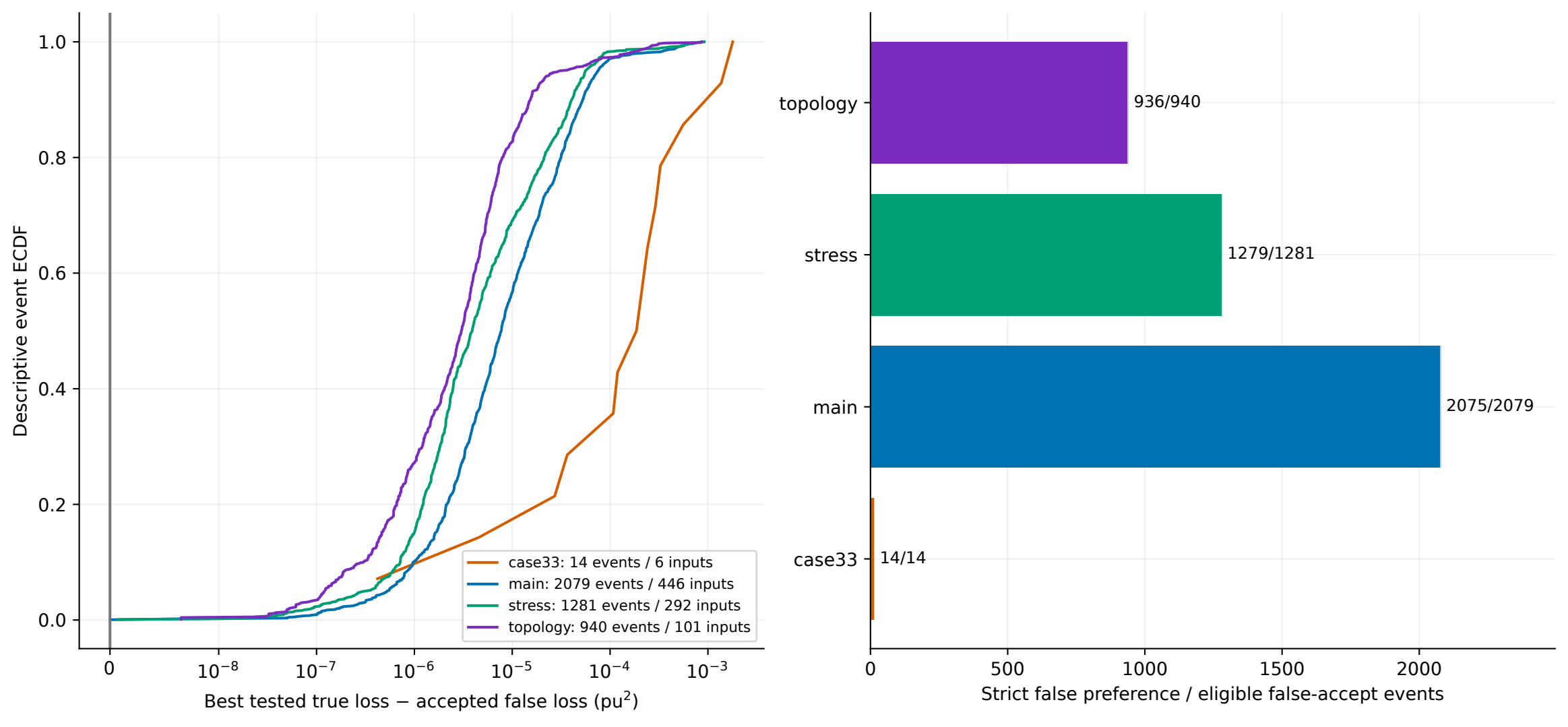

Delete after two-step − two-step
Two-step + delete − delete only
Two-step − single-step
0.00
0.01
0.02
0.03
0.04
0.05
10 bundle means and 95% mean-gain interval

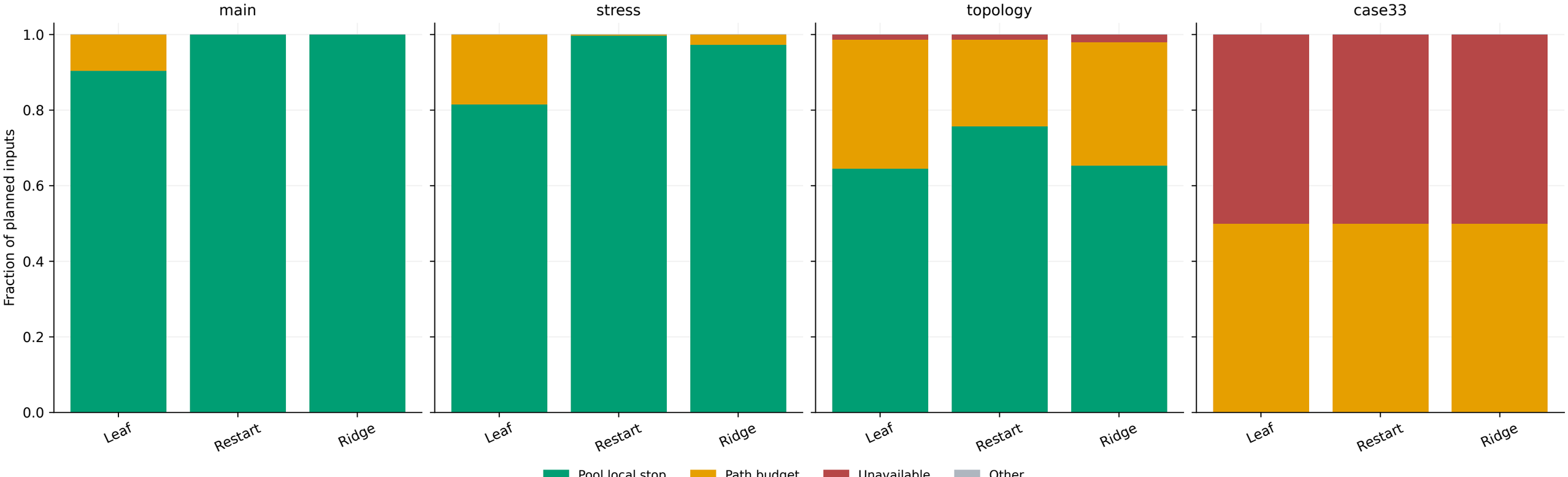

Delete after two-step − two-step
26
131
3
Two-step + delete − delete only
5
154
1
Two-step − single-step
5
154
1
0
20
40
60
80
100
120
140
160
Inputs: improve / tie / degrade (160 per comparison)

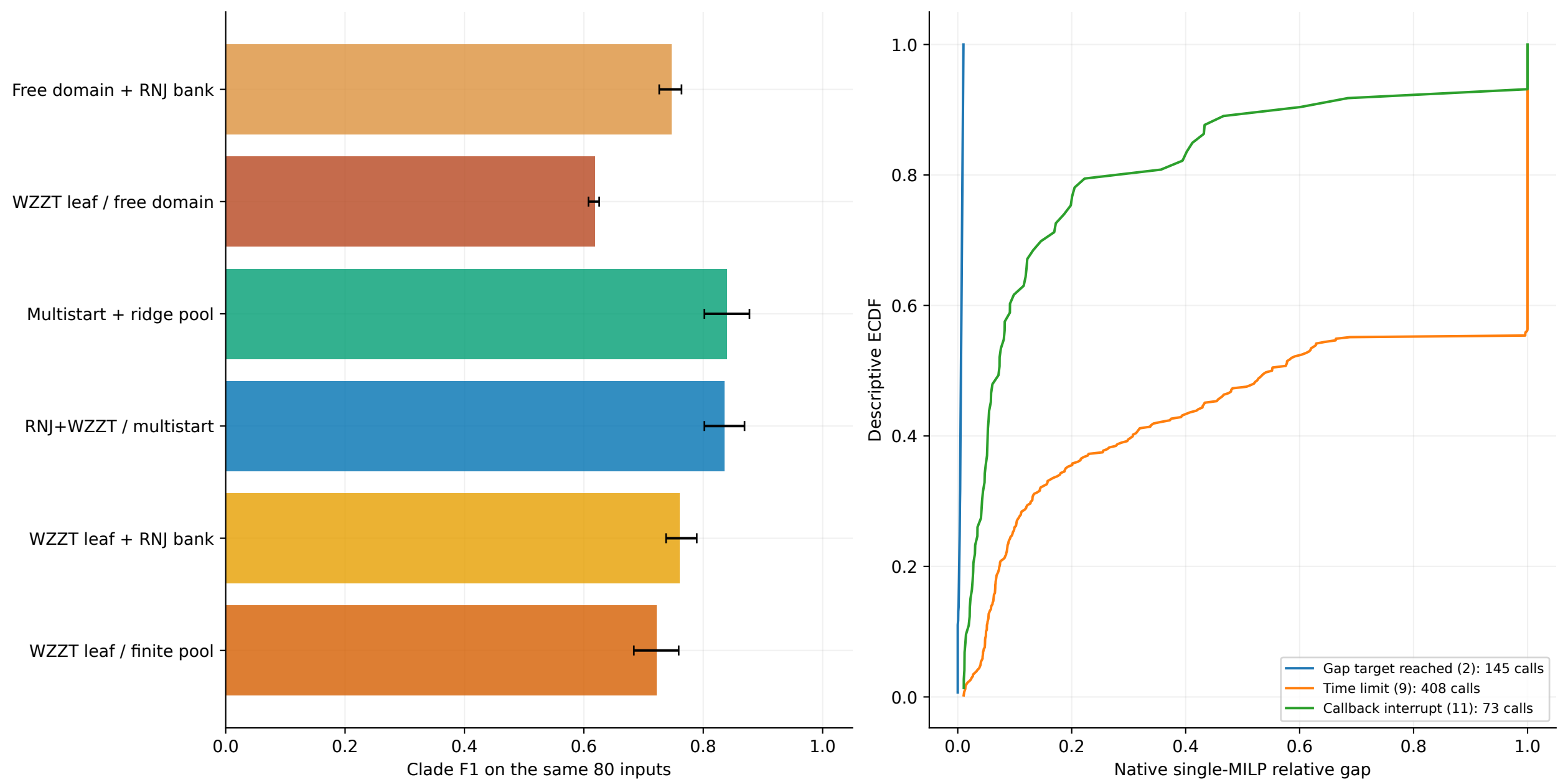

main | 640 inputs
stress | 400 inputs
topology | 144 inputs
case33 | 20 inputs
Mean clade F1
Available-model test MAE (pu²)
Final strength floor

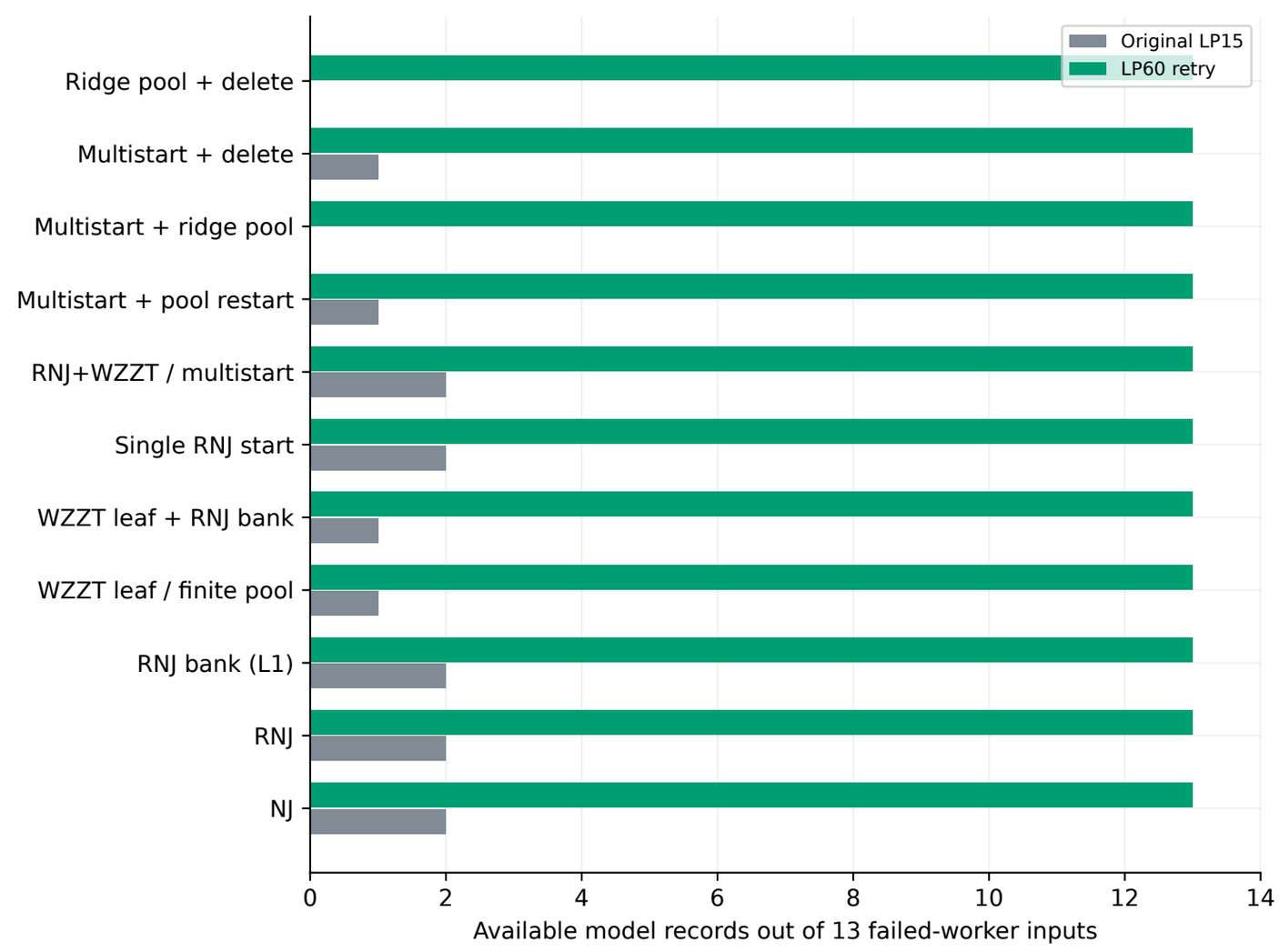

main: 640 available inputs
stress: 400 available inputs
topology: 141 available inputs
case33: 10 available inputs
Guard minus fixed-floor F1
Guard minus fixed-floor validation MAE (pu²)

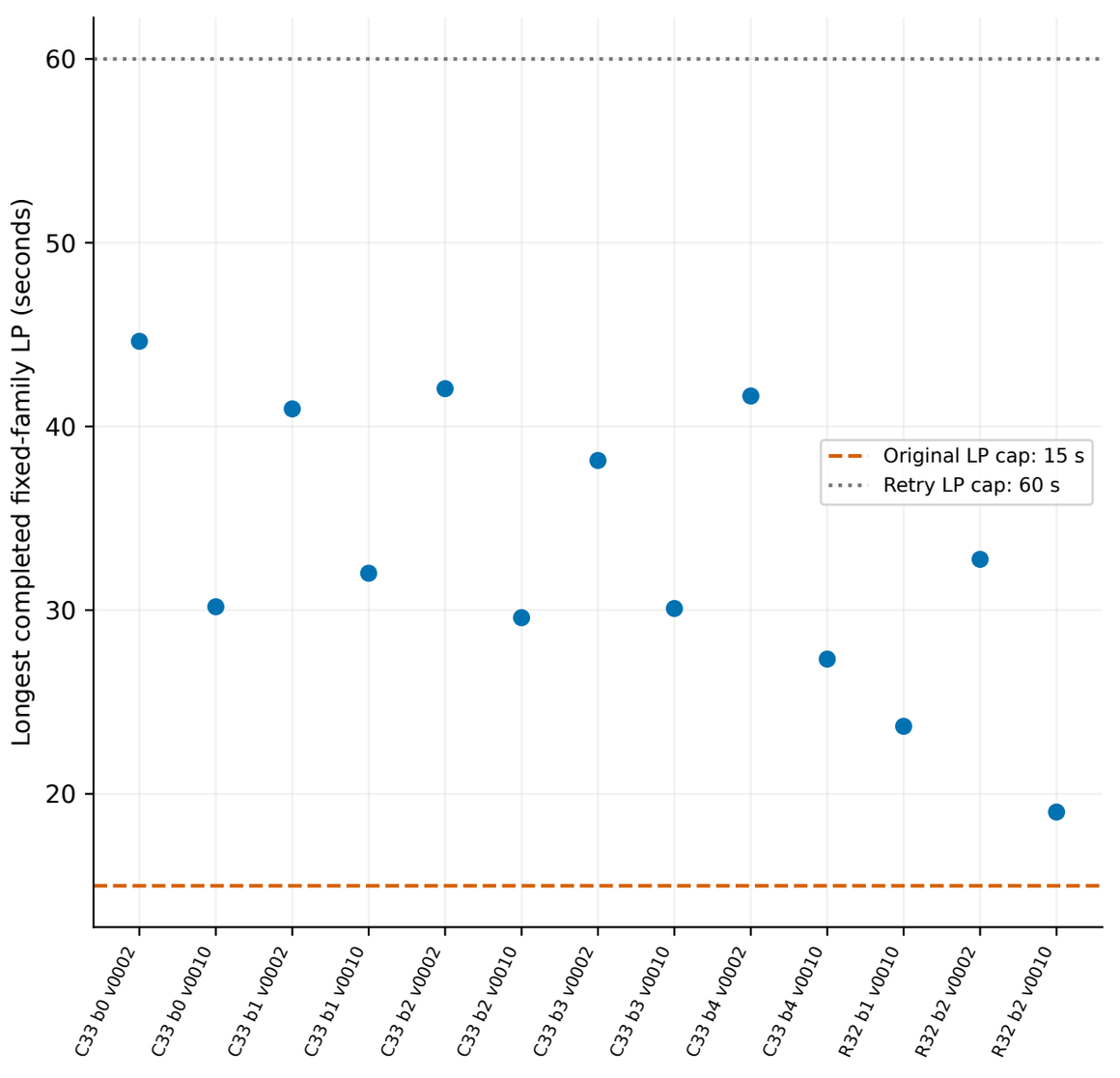

main
stress
topology
case33
Fraction of inputs
1.0
0.8
0.6
0.4
0.2
0.0
Pool miss
Crossing
Overfine
Unavailable
RNJ+WZZT / multistart
Multistart + ridge pool
Ridge pool + delete

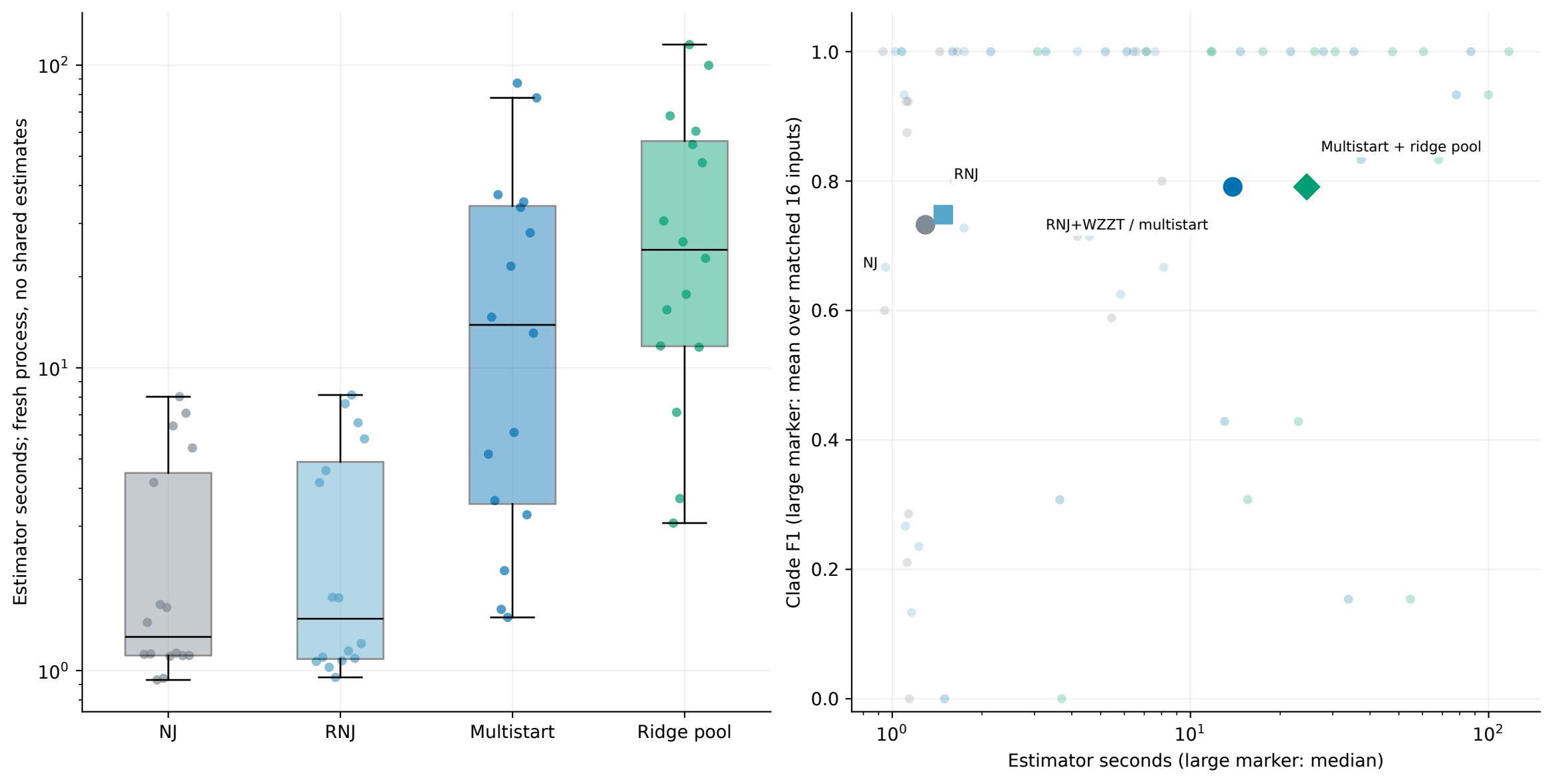

Descriptive event ECDF
1.0
0.8
0.6
0.4
0.2
0.0
0
$10^{-8}$
$10^{-7}$
$10^{-6}$
$10^{-5}$
$10^{-4}$
$10^{-3}$
Best tested true loss − accepted false loss (pu$^2$)
case33: 14 events / 6 inputs
main: 2079 events / 446 inputs
stress: 1281 events / 292 inputs
topology: 940 events / 101 inputs
topology
stress
main
case33
936/940
1279/1281
2075/2079
14/14
0
500
1000
1500
2000
Strict false preference / eligible false-accept events

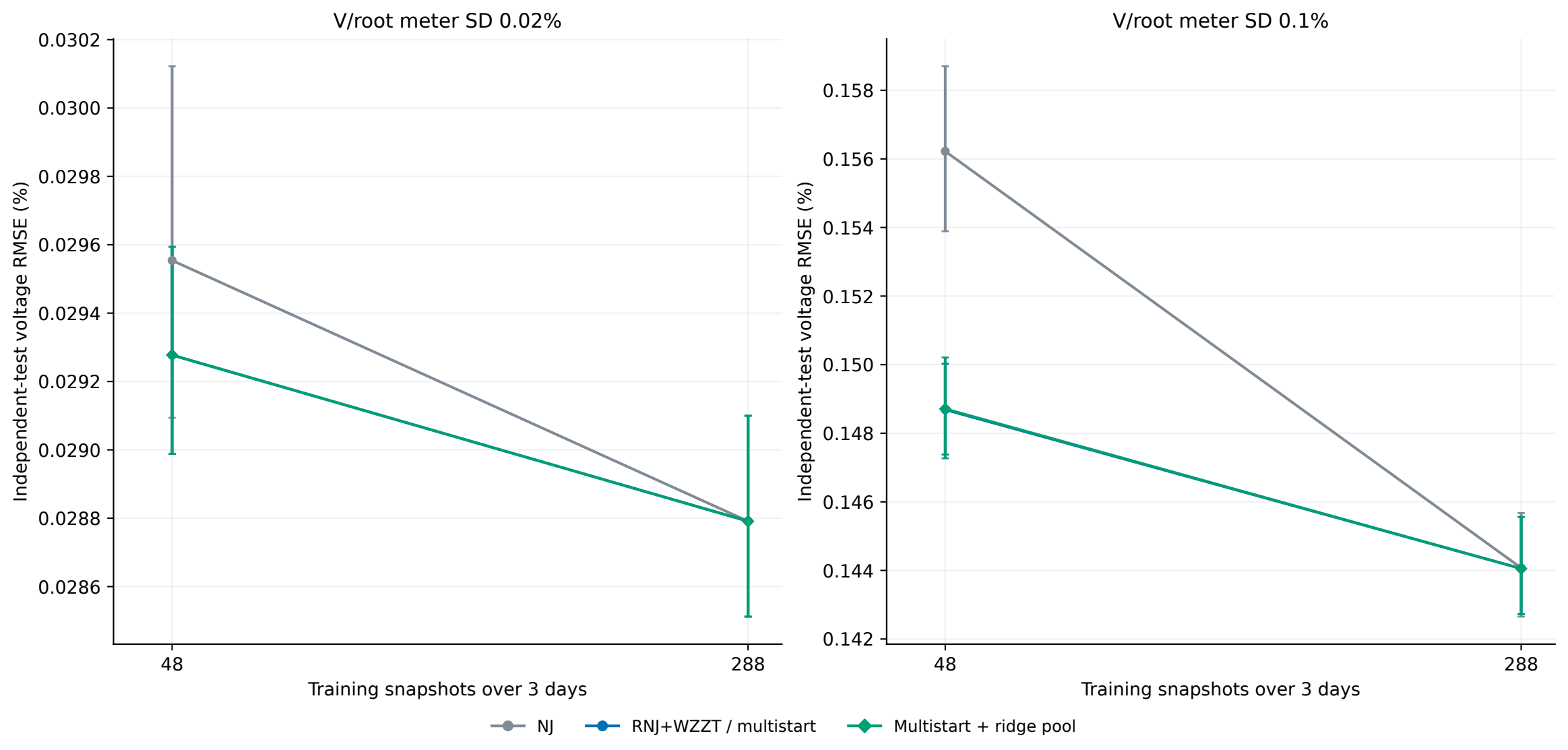

main
stress
topology
case33
Fraction of planned inputs
1.0
0.8
0.6
0.4
0.2
0.0
Leaf
Restart
Ridge
Pool local stop
Path budget
Unavailable
Other

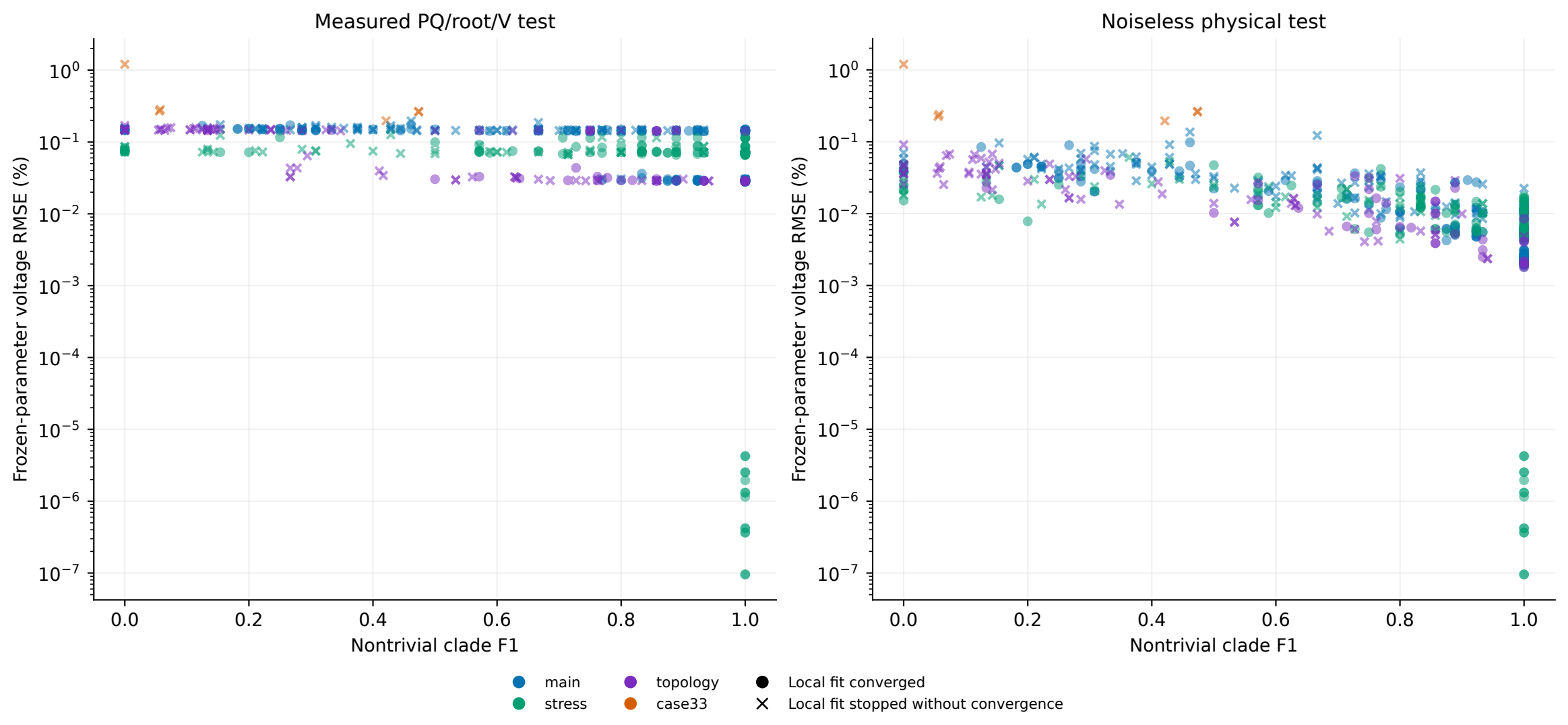

Free domain + RNJ bank
WZZT leaf / free domain
Multistart + ridge pool
RNJ+WZZT / multistart
WZZT leaf + RNJ bank
WZZT leaf / finite pool
0.0
0.2
0.4
0.6
0.8
1.0
Clade F1 on the same 80 inputs
Descriptive ECDF
Native single-MILP relative gap
Gap target reached (2): 145 calls
Time limit (9): 408 calls
Callback interrupt (11): 73 calls

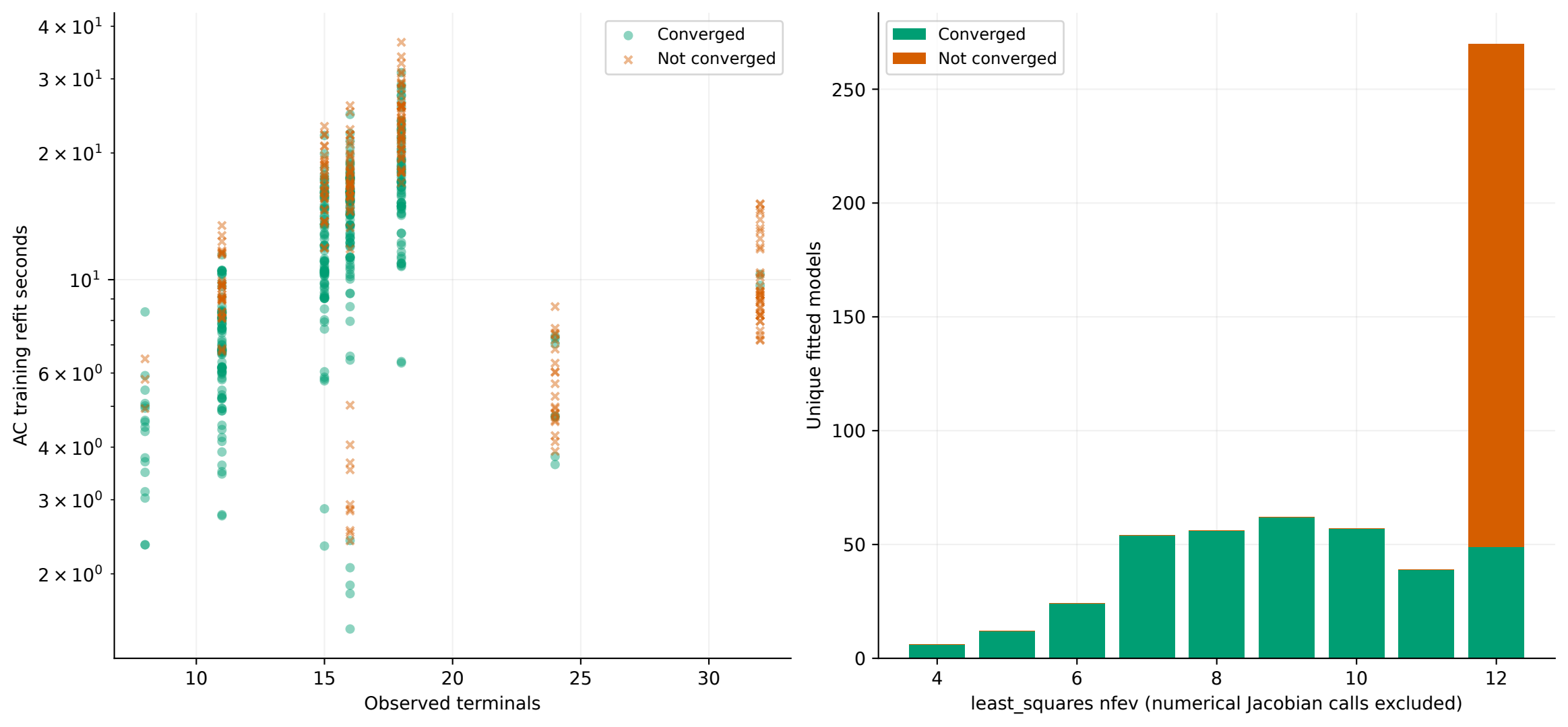

Original LP15
LP60 retry
Ridge pool + delete
Multistart + delete
Multistart + ridge pool
Multistart + pool restart
RNJ+WZZT / multistart
Single RNJ start
WZZT leaf + RNJ bank
WZZT leaf / finite pool
RNJ bank (L1)
RNJ
NJ
0
2
4
6
8
10
12
14
Available model records out of 13 failed-worker inputs


Longest completed fixed-family LP (seconds)
60
50
40
30
20
Original LP cap: 15 s
Retry LP cap: 60 s
C33 b0 v0002
C33 b0 v0010
C33 b1 v0002
C33 b1 v0010
C33 b2 v0002
C33 b2 v0010
C33 b3 v0002
C33 b3 v0010
C33 b4 v0002
C33 b4 v0010
R32 b1 v0010
R32 b2 v0002
R32 b2 v0010


Estimator seconds; fresh process, no shared estimates
$10^2$
$10^1$
$10^0$
NJ
RNJ
Multistart
Ridge pool
Clade F1 (large marker: mean over matched 16 inputs)
1.0
0.8
0.6
0.4
0.2
0.0
Multistart + ridge pool
RNJ
RNJ+WZZT / multistart
NJ
$10^0$
$10^1$
$10^2$
Estimator seconds (large marker: median)

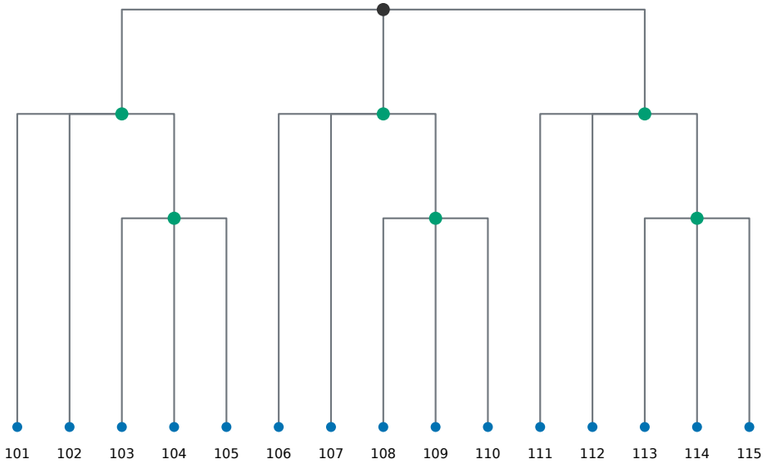

V/root meter SD 0.02%
V/root meter SD 0.1%
Independent-test voltage RMSE (%)
0.0302
0.0300
0.0298
0.0296
0.0294
0.0292
0.0290
0.0288
0.0286
0.158
0.156
0.154
0.152
0.150
0.148
0.146
0.144
0.142
48
288
Training snapshots over 3 days
NJ
RNJ+WZZT / multistart
Multistart + ridge pool


Measured PQ/root/V test
Noiseless physical test
Frozen-parameter voltage RMSE (%)
Nontrivial clade F1
0.0
0.2
0.4
0.6
0.8
1.0
$10^{0}$
$10^{-1}$
$10^{-2}$
$10^{-3}$
$10^{-4}$
$10^{-5}$
$10^{-6}$
$10^{-7}$
main
stress
topology
case33
Local fit converged
Local fit stopped without convergence

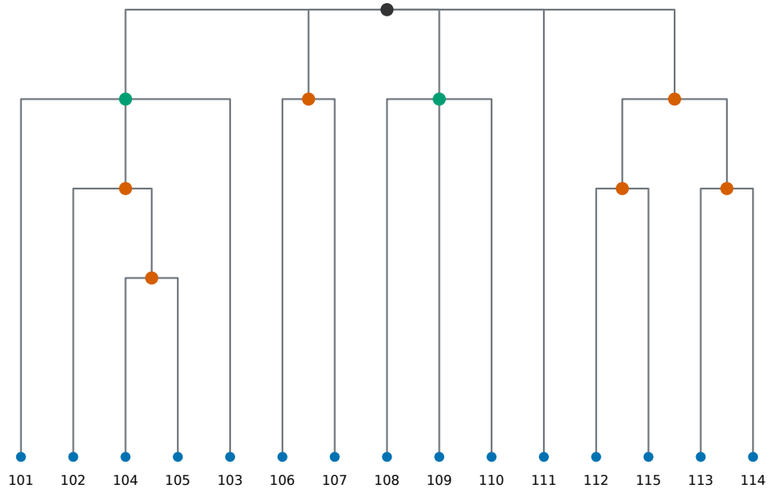

Converged
Not converged
AC training refit seconds
$4 \times 10^1$
$3 \times 10^1$
$2 \times 10^1$
$10^1$
$6 \times 10^0$
$4 \times 10^0$
$3 \times 10^0$
$2 \times 10^0$
10
15
20
25
30
Observed terminals
Converged
Not converged
Unique fitted models
250
200
150
100
50
0
4
6
8
10
12
least_squares nfev (numerical Jacobian calls excluded)

# paper15 | bundle 0 | N=48 | V/root SD=0.1% | original LP15

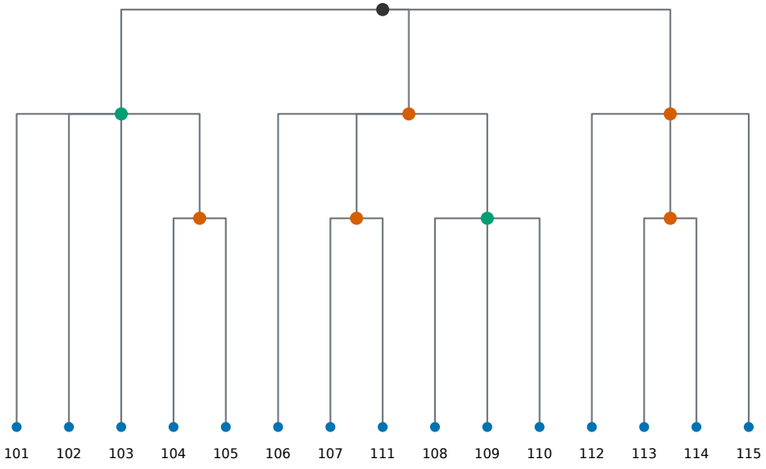
True canonical clade tree



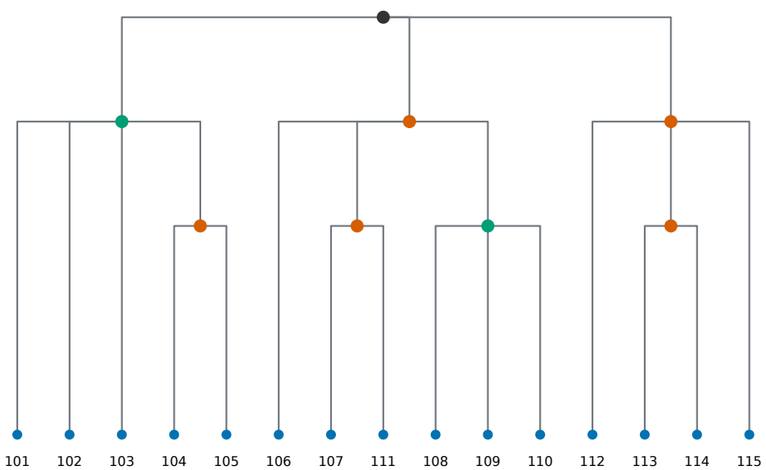

NJ
F1=0.286 | TP 2, FP 6, FN 4



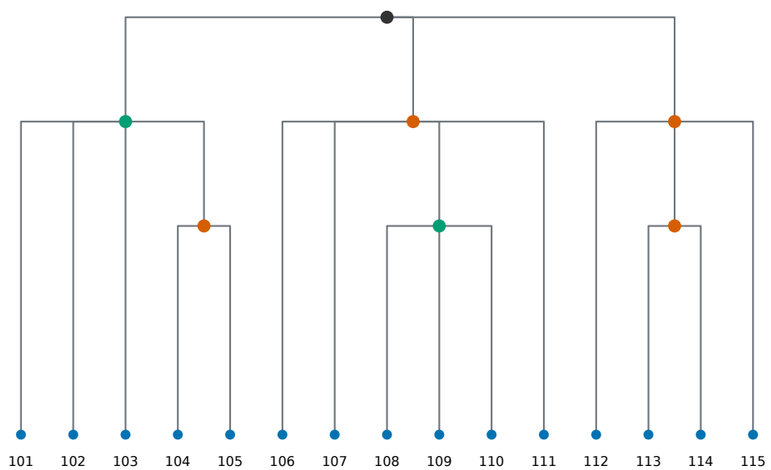

RNJ+WZZT / multistart
F1=0.308 | TP 2, FP 5, FN 4




Multistart + ridge pool
F1=0.308 | TP 2, FP 5, FN 4




Ridge pool + delete
F1=0.333 | TP 2, FP 4, FN 4



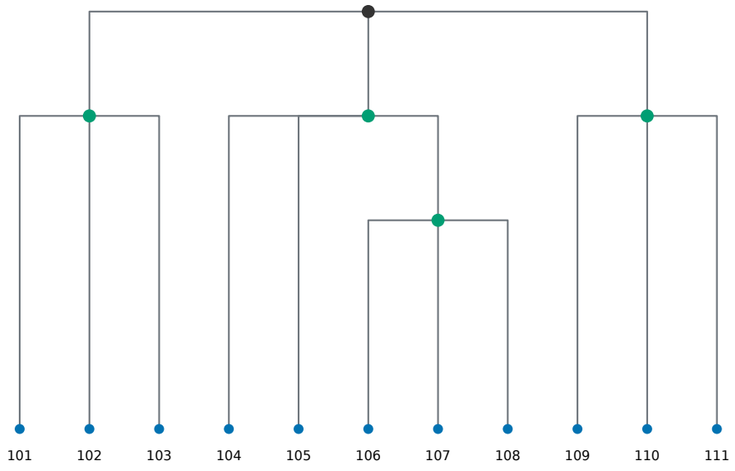

Canonical clade trees

Blue: observed terminal
Green: correct nontrivial clade
Orange: false nontrivial clade
Black: root

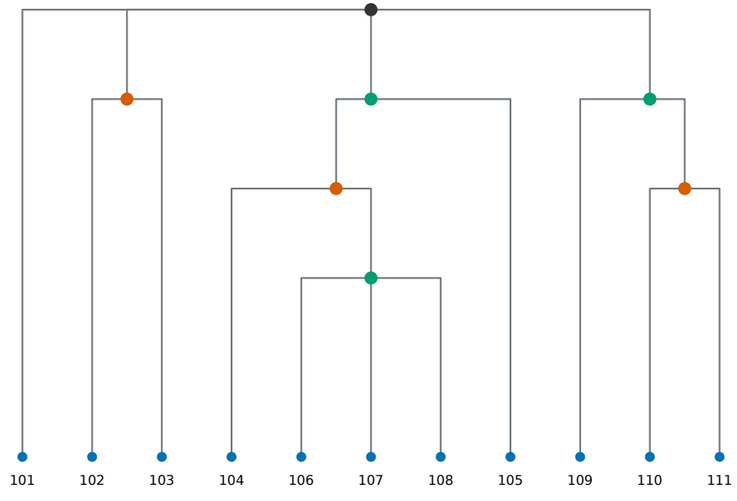
Edges show hierarchy; lengths are layout only.
Terminal order follows each tree traversal.

soumalas11 | bundle 0 | N=48 | V/root SD=0.1% | original LP15

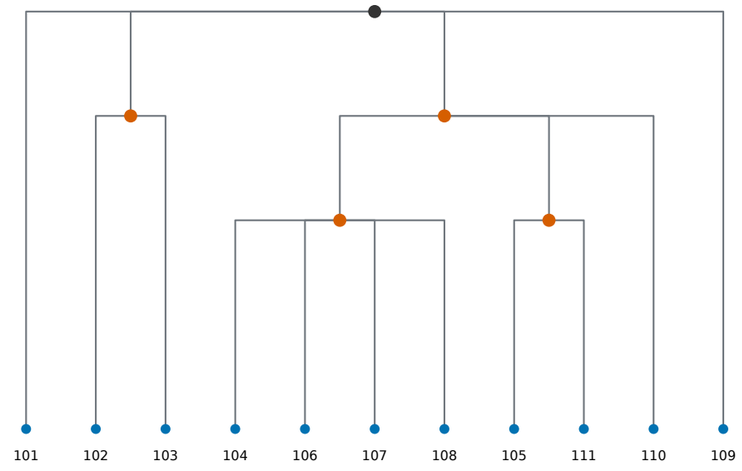
True canonical clade tree



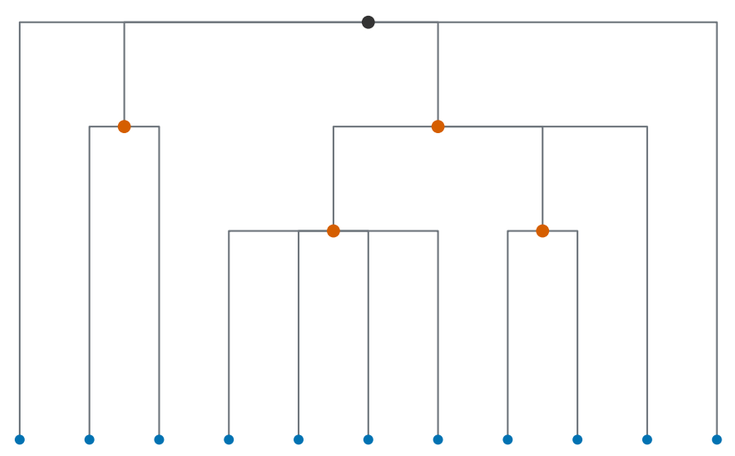

NJ
F1=0.600 | TP 3, FP 3, FN 1



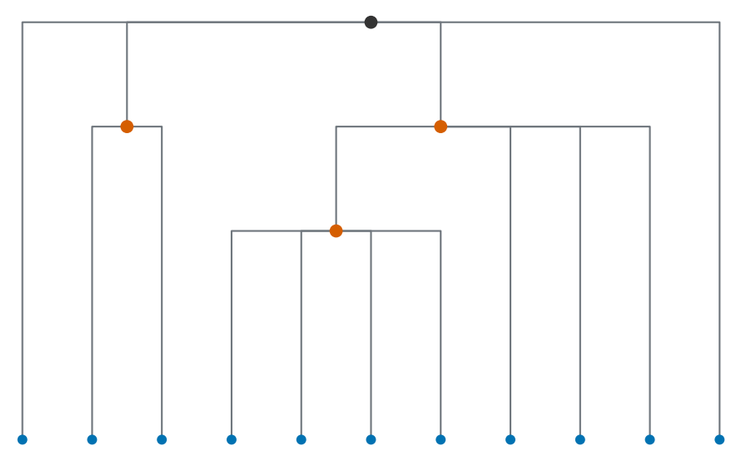

RNJ+WZZT / multistart
F1=0.000 | TP 0, FP 4, FN 4




Multistart + ridge pool
F1=0.000 | TP 0, FP 4, FN 4




Ridge pool + delete
F1=0.000 | TP 0, FP 3, FN 4



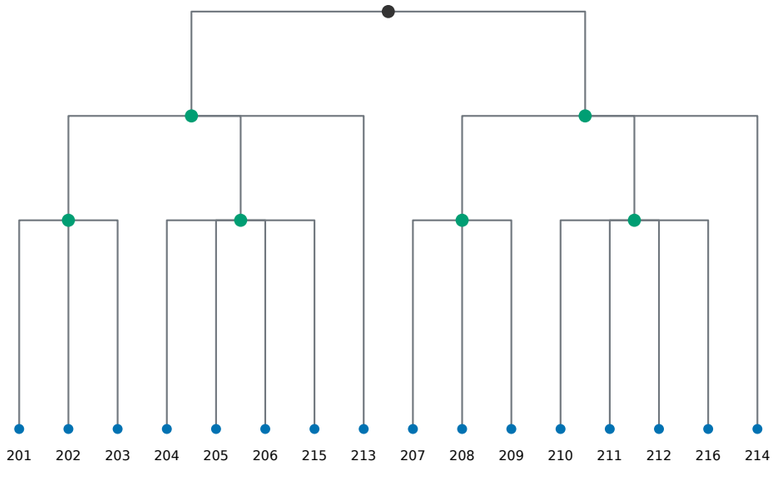

Canonical clade trees

Blue: observed terminal
Green: correct nontrivial clade
Orange: false nontrivial clade
Black: root

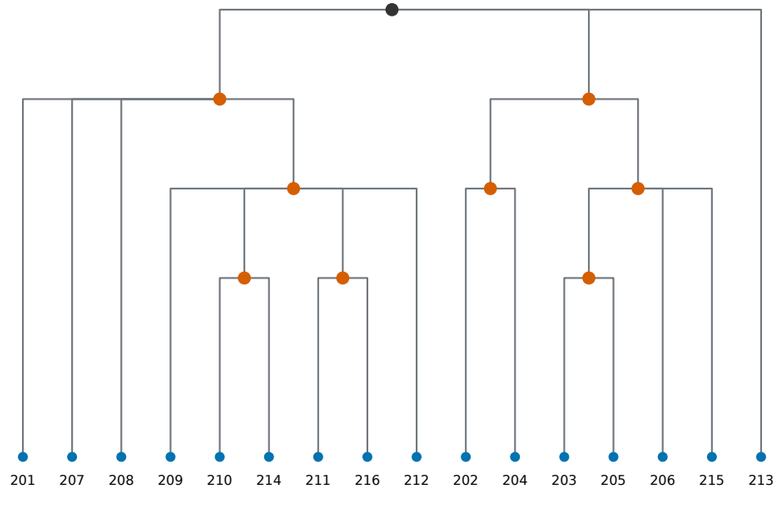
Edges show hierarchy; lengths are layout only.
Terminal order follows each tree traversal.

# flynn16 | bundle 0 | N=48 | V/root SD=0.1% | original LP15

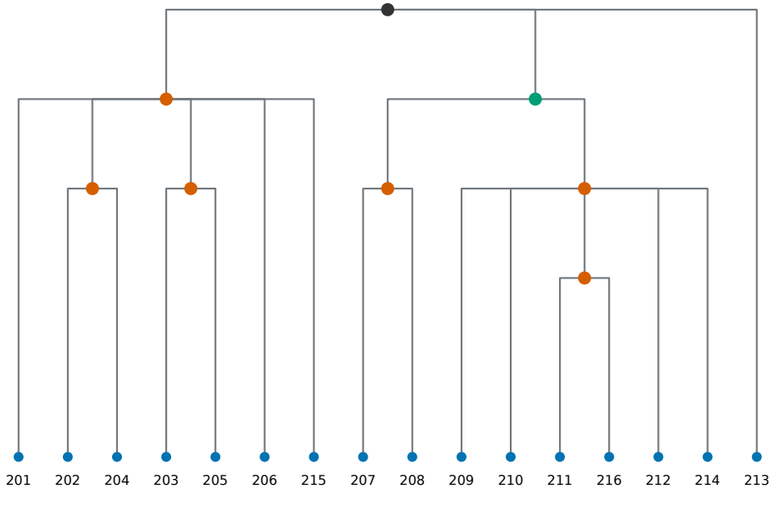
True canonical clade tree



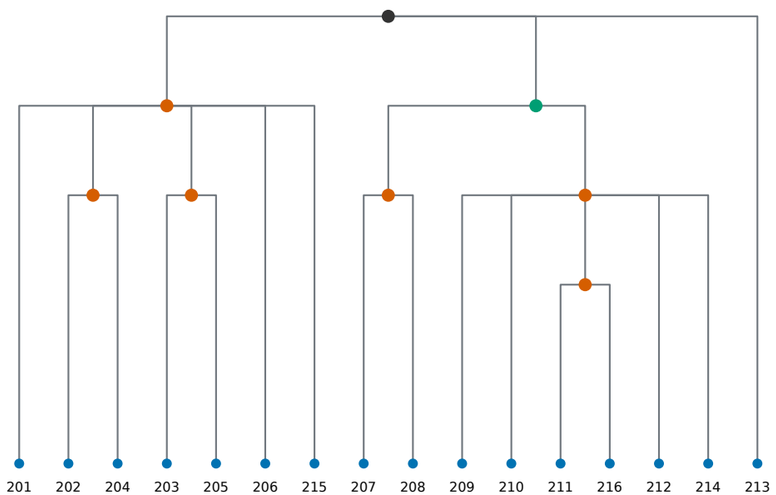

NJ
F1=0.000 | TP 0, FP 8, FN 6



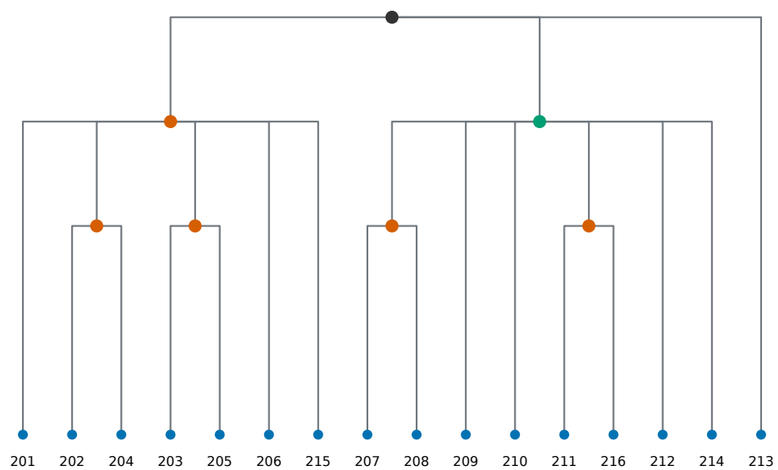

RNJ+WZZT / multistart
F1=0.154 | TP 1, FP 6, FN 5




Multistart + ridge pool
F1=0.154 | TP 1, FP 6, FN 5




Ridge pool + delete
F1=0.167 | TP 1, FP 5, FN 5



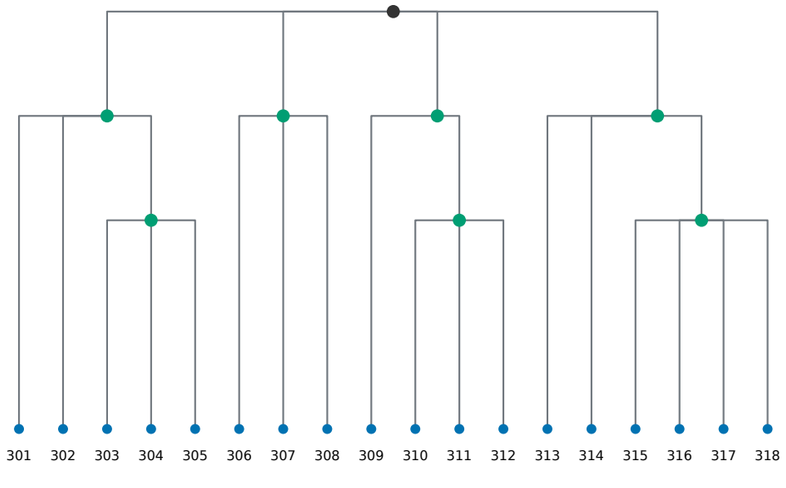

Canonical clade trees

Blue: observed terminal
Green: correct nontrivial clade
Orange: false nontrivial clade
Black: root

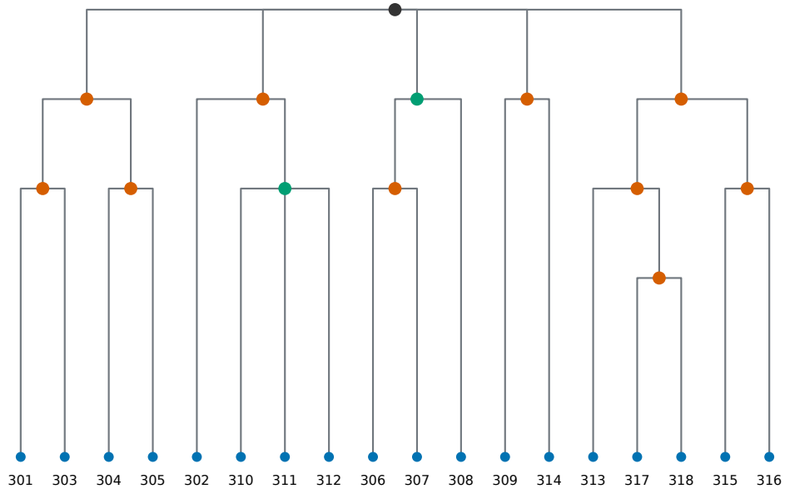
Edges show hierarchy; lengths are layout only.
Terminal order follows each tree traversal.

# pengwah18 | bundle 0 | N=48 | V/root SD=0.1% | original LP15

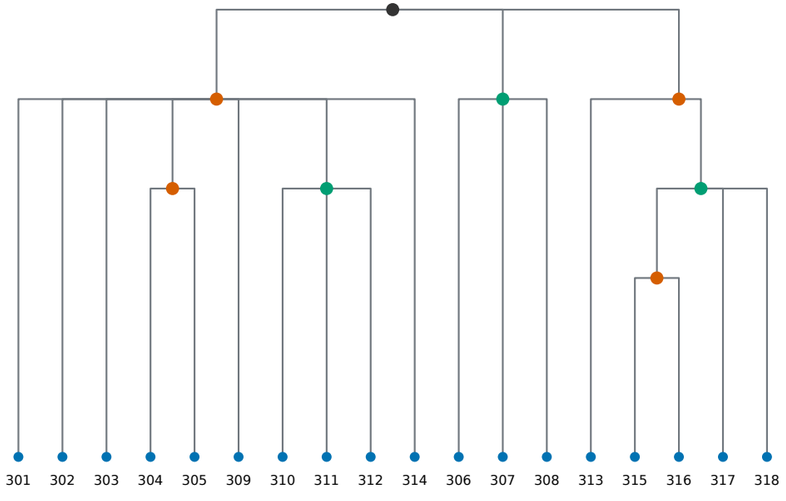
True canonical clade tree



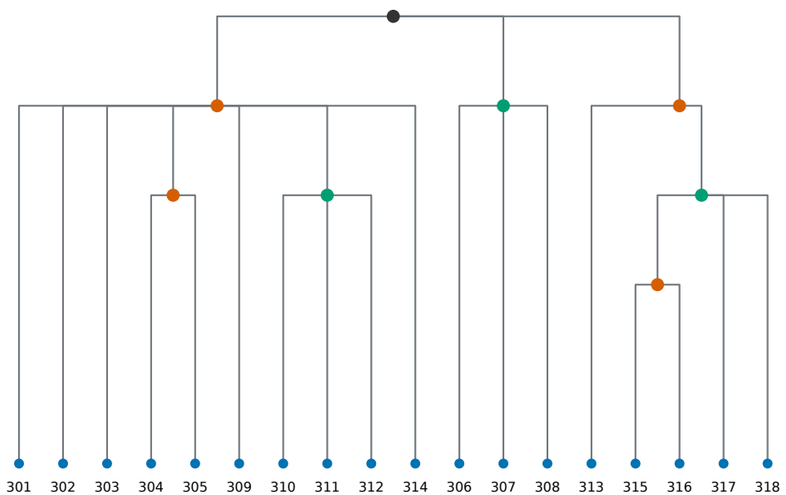

NJ
F1=0.211 | TP 2, FP 10, FN 5



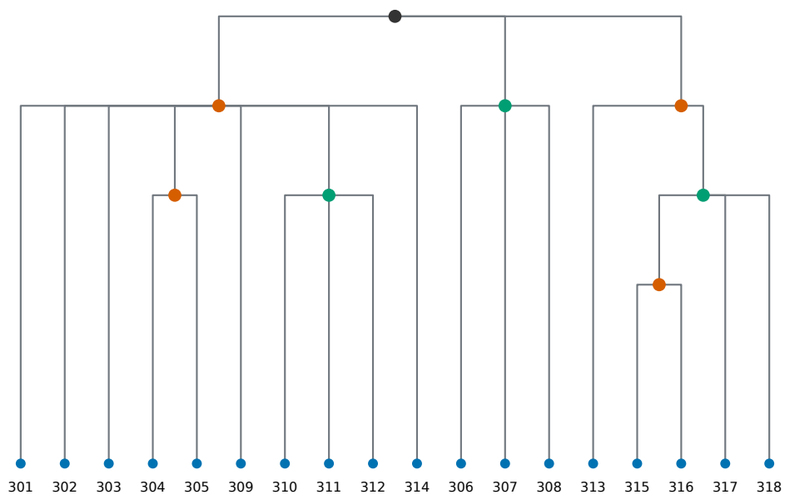

RNJ+WZZT / multistart
F1=0.429 | TP 3, FP 4, FN 4




Multistart + ridge pool
F1=0.429 | TP 3, FP 4, FN 4




Ridge pool + delete
F1=0.429 | TP 3, FP 4, FN 4



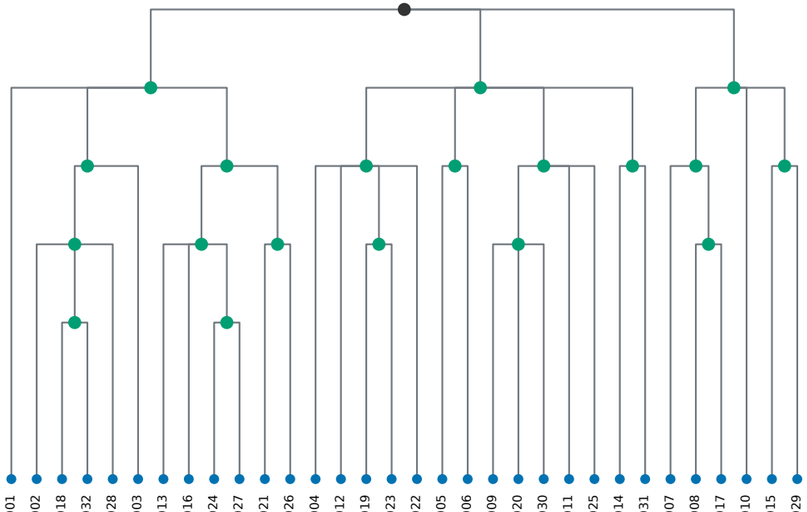

Canonical clade trees

Blue: observed terminal
Green: correct nontrivial clade
Orange: false nontrivial clade
Black: root

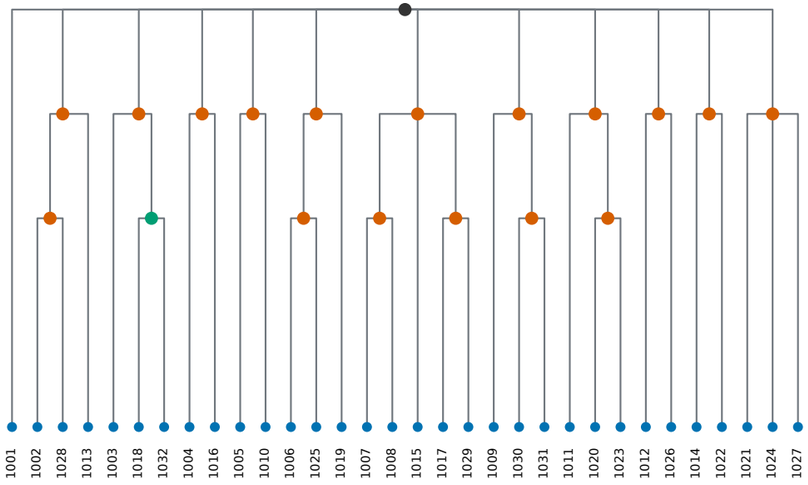
Edges show hierarchy; lengths are layout only.
Terminal order follows each tree traversal.

random32_t1 | bundle 0 | N=288 | V/root SD=0.1% | original LP15

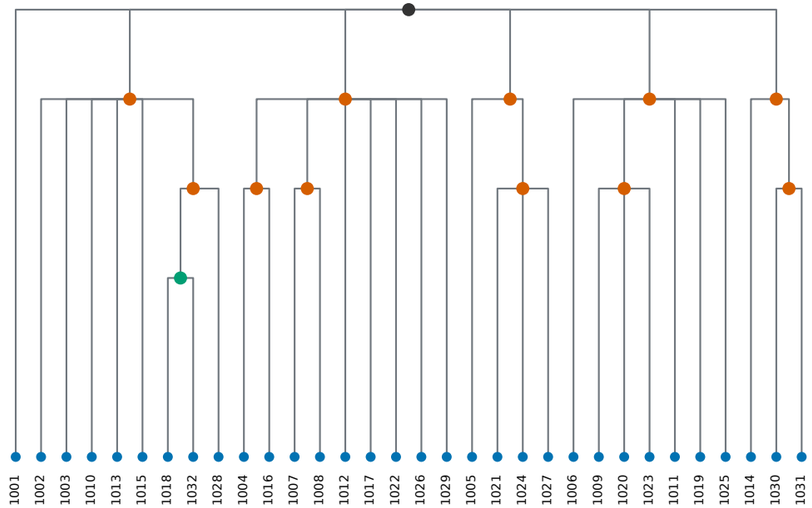
True canonical clade tree



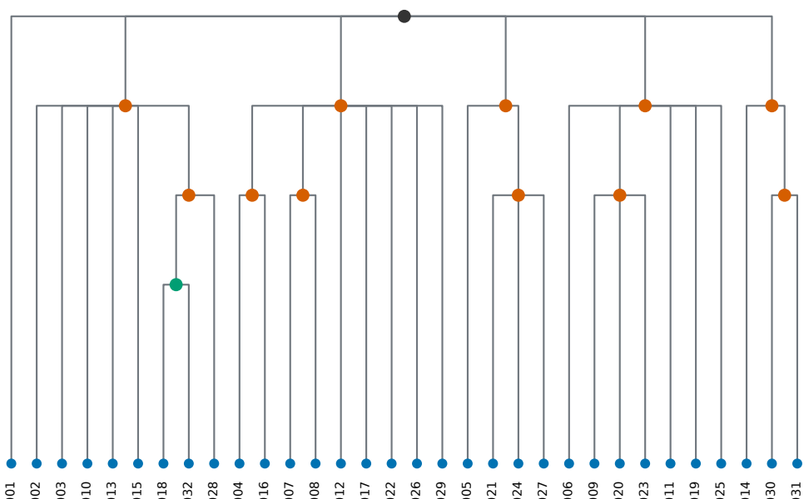

NJ
F1=0.054 | TP 1, FP 17, FN 18



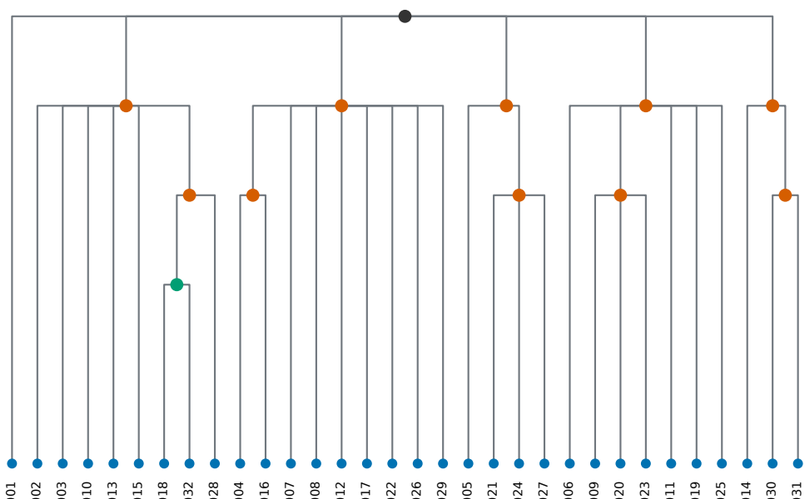

RNJ+WZZT / multistart
F1=0.065 | TP 1, FP 11, FN 18




Multistart + ridge pool
F1=0.065 | TP 1, FP 11, FN 18




Ridge pool + delete
F1=0.067 | TP 1, FP 10, FN 18



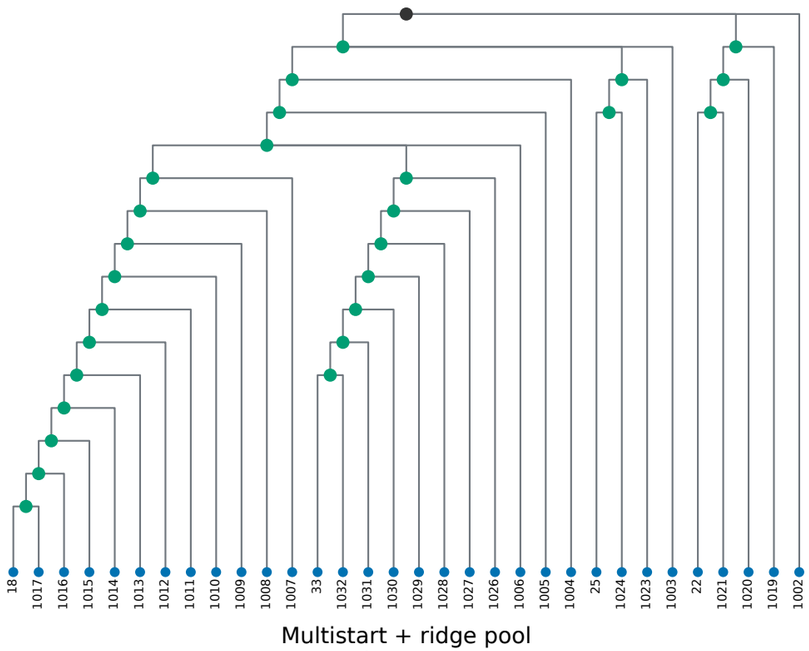

Canonical clade trees

Blue: observed terminal
Green: correct nontrivial clade
Orange: false nontrivial clade
Black: root

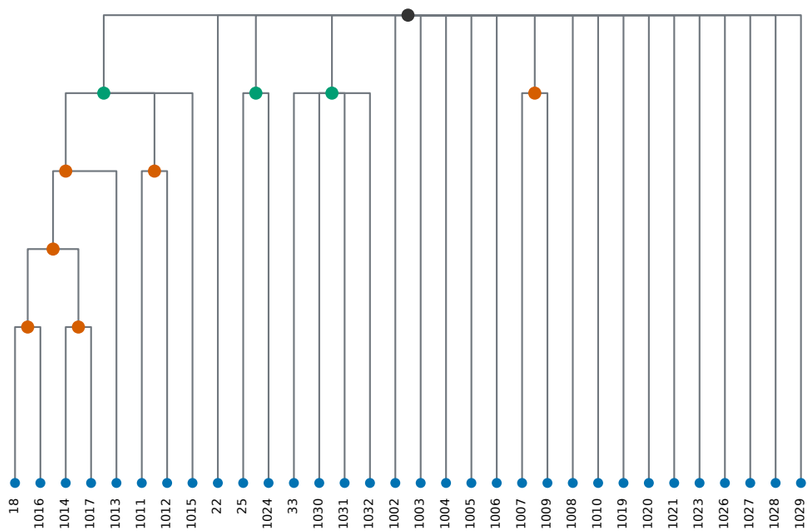
Edges show hierarchy; lengths are layout only.
Terminal order follows each tree traversal.

# case33_terminal | bundle 0 | N=288 | V/root SD=0.1% | LP60 retry

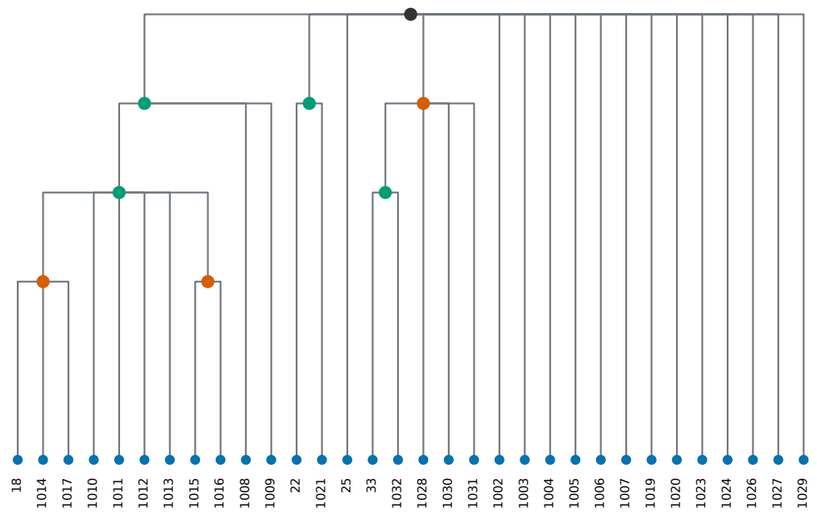
True canonical clade tree



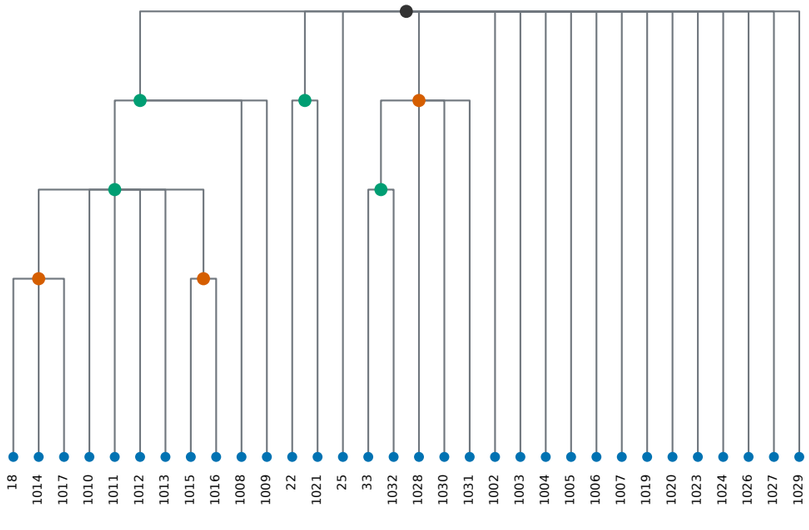

NJ
F1=0.167 | TP 3, FP 6, FN 24



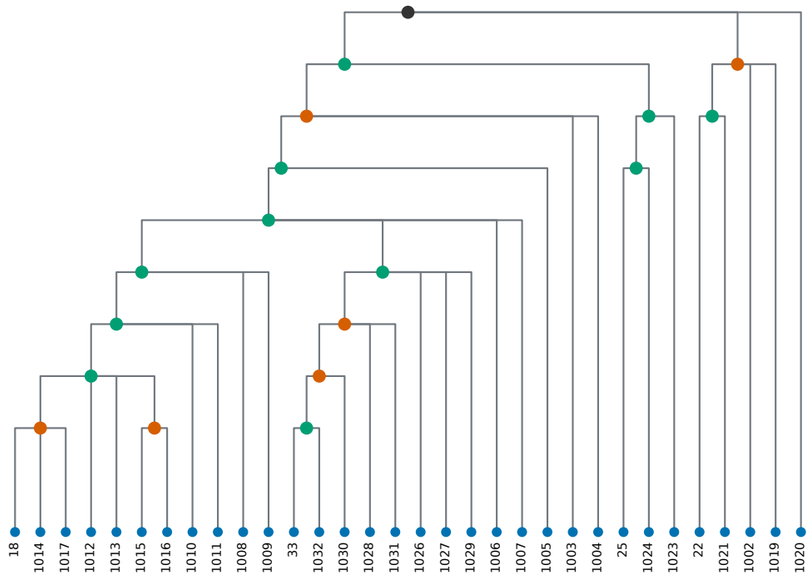

RNJ+WZZT / multistart
F1=0.235 | TP 4, FP 3, FN 23




Multistart + ridge pool
F1=0.235 | TP 4, FP 3, FN 23




Validation filter guard
F1=0.500 | TP 11, FP 6, FN 16




Canonical clade trees

Blue: observed terminal
Green: correct nontrivial clade
Orange: false nontrivial clade
Black: root

Edges show hierarchy; lengths are layout only.
Terminal order follows each tree traversal.